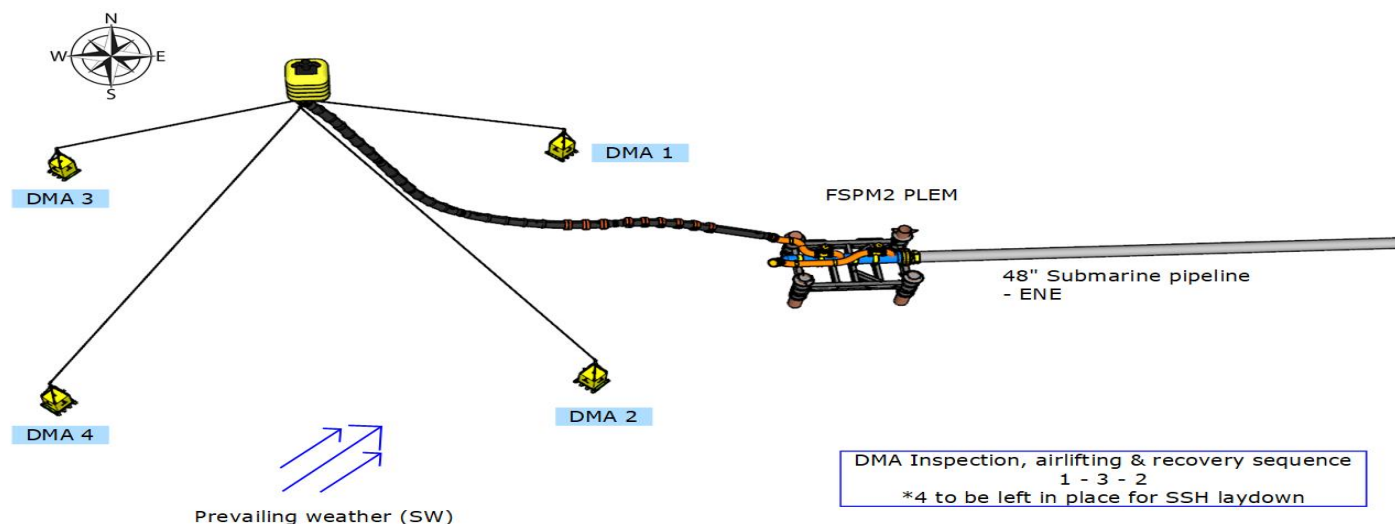




ACOE Temporary Export System Decommissioning Project



ACOE Decommission – PLAN: -SITE EMERGENCY RESPONSE PLAN

Project: Decommissioning of Forcados ACOE Temporary Export System

Client: Shell Petroleum Development Company of Nigeria (SPDC)

Contractor: Century Ports & Terminals LTD (CPTL)

Subcontractor: HydroDive Nigeria Limited

Project No.: 863-01-24

Document: Forcados SPM 2 – Plan-Site Emergency Response Plan

Document No.: HDG-PRJ-HSE-863-01-24

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Record of Revision							
Revision Code	Issue Date	Reason for Issue	Prepared	Reviewed	Approved		SPDC
					HSE Manager	Diving TA	
			Nick Roddy	D. Golding	Kene Anyabolu	Nick Roddy	David Mooney
A01	24/01/2025	Approved for use	N. Roddy	D. Golding	K. Anyabolu	N. Roddy	D. Mooney
R03	23/01/2025	Issued for review.	N. Roddy	D. Golding	K. Anyabolu	N. Roddy	D. Mooney
R02	22/01/2025	Issued for review.	N. Roddy	D. Golding	K. Anyabolu	N. Roddy	D. Mooney
R01	20/01/2025	Issued for review.	N. Roddy	D. Golding	K. Anyabolu	N. Roddy	D. Mooney

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Party	Ref Ind.	Name	Sign	Date

Revision Philosophy:

- Revisions for review shall be issued at R01, with subsequent R02, R03, etc., as required.
- All revisions approved for issue or design shall be issued at A01, with subsequent A02, A03, etc as required.
- Documents approved for Construction shall be issued at C01, C02, C03, etc. as required.
- Documents or drawings revised as “As built” shall be issued as Z01, Z02, Z03, etc. as required.

Narrative sections revised from previous approved issues are to be noted in the table below and/or highlighted in the RH margin using the appropriate revision status, thus: A02.

- Previous revision highlighting to be removed at subsequent issues.

Drawings/diagrams revised from previous approved issues are highlighted by 'clouding' the affected areas and using a triangle containing the revision status.

If a document is cancelled the document shall be issued at the next sequential number and suffixed with an X. For example, if 02R is the current revision then 03X shall be the Cancelled revision.

Revision History				
Date	Section	Reason for Change	Document No.	Revision No.
20/01/2025	All	Issued for review	HDG-PRJ-HSE-863-01-24	R01
22/01/2025	All	Issued for review	HDG-PRJ-HSE-863-01-24	R02
23/01/2025	All	Issued for review	HDG-PRJ-HSE-863-01-24	R03
24/01/2025	All	Approved for use	HDG-PRJ-HSE-863-01-24	A01

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1 Objective

The work that HydroDive performs for its customers is in a harsh and dangerous environment where when emergency situations present themselves, it is imperative then that there is in place a logical process that must be followed strictly so as to prevent further harm, save life and stop/limit damage to property and the environment.

It must be recognized that most emergencies cannot be handled alone and as such there needs to be a system whereby additional internal and external resources should be available at times of an emergency.

This procedure covers the activities to be performed whenever an emergency arises “24/7”

2 Brief Overview Of The Decommissioning Of The Temporary Export System

Century Port & Terminals LTD (CPTL) has been awarded a contract by Shell Petroleum Development Company of Nigeria (SPDC) to decommission and remove a temporary ACOE Export System at Forcados SPM 2 location, in the Forcados Field, 29.3 km offshore Nigeria. The completion of the diving scope has been awarded to Century Ports and Terminals Limited. Prior to installation, the buoy presently occupying the location was removed by Hydrodive. HydroDive (Subcontractor/Company) has been contracted by CPTL (Contractor), on behalf of Shell (Client), to provide subsea diving support services for all works associated with the temporary export system decommissioning.

A Procedure has been developed to describe the way the diving scope of work will be executed. All diving works will be performed from the Dive Support Vessels, and DSV HD Contender.

HydroDive Nigeria Limited is the owner of the and HD Contender and as such all diving operations will be conducted in strict accordance with HydroDive’s procedures and policies. Diving Operations Manual, Doc Number HDG-DIV-MAN-001 and Health Safety & Environmental Manual Doc number HDG-HSE-MAN-00 (1&2) and International Marine Contractors Association (IMCA) industry guidance, specifically to IMCA D014 Code of Practice for Offshore Diving.

2.1 Project Location

The Forcados SPM 1 Buoy (SO-1207) is located just offshore 29.3km from the Forcados terminal, in water depth ranging from 20m to 25 m. The CALM buoy berth is approximately moored 1.54km away from the Crude Loading Platform.

The Forcados SPM 2 Buoy (SO-12481) is located just offshore 28km from the Forcados terminal, in water depth ranging from 20m to 25 m. The CALM buoy berth is approximately moored 1.54km away from the Crude Loading Platform.

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3 HYDRODIVE EMERGENCY RESPONSE POLICY



EMERGENCY RESPONSE POLICY STATEMENT

It is the policy of HydroDive Nigeria Limited in the conduct of its operations; priority will be given above all other considerations, to:

- Proactively implement a structured HSE Management System that prevents emergencies; and
- Manage all emergencies to enable restoration of normalcy within the shortest possible timeframe.

Our goal is to:

- Save life;
- Care for the injured;
- Protect the environment;
- Minimize damage to assets;
- Defend HydroDive's good corporate image and reputation;
- Safely stabilize and restore normal operations; and
- Learn from emergencies to prevent reoccurrence.

We shall ensure that an operation is never so important that it cannot be carried out safely and with due care and attention to prevent emergencies. In carrying out this intention, the following principles should be applied:

- Set-up appropriate response organization, facilities, procedures and competent personnel to coordinate actions promptly to effectively manage any emergency situation;
- Accord emergency response precedence over normal duty;
- Maintain close liaison with government, industry organizations and other relevant stakeholders while resolving the emergency;
- Ensure availability of emergency response facilities at all times;
- Ensure emergency response equipment are maintained and accorded high priority;
- Conduct periodic drills to verify effectiveness of emergency response systems;
- Provide injured parties affected by emergency arising from HydroDive's operations with appropriate medical attention until they are reasonably stable; and
- Require its contractors to manage emergency in their operations in a manner that is consistent with this policy.


Frank Ifedi
 Managing Director
 HydroDive Group

HDG-HSE-POL-011_A07_14th January 2025

4 SCOPE OF THIS DOCUMENT

This document covers emergency response from the Hydrodive vessel HD Contender

IMO number	9122796	
MMSI	657939000	
Helideck Class H2	Helicopters from 15 m up to but not including 24 m	

This procedure provides a Table in the event of an emergency with various contact numbers and emails addresses (See **Appendix 1**. It also provides various emergency scenarios and how they should be handled (See Section 12 of this procedure).

The document endeavours to provide a robust frame work that can deal with all reasonably foreseeable events and is solid enough to adapt to others where necessary.

This document sets out HydroDive's policy with regards to how personnel should proceed when confronted with an emergency while during offshore operations and an overview of the HydroDive's Emergency Response Organization and the co-operation expected and required. Any emergency involving chartered vessels or property damage or requiring airborne medical evacuation will involve close cooperation with SPDC Emergency Services.

This procedure forms the basis of a project specific bridging document, which shall incorporate SPDC emergency response communications procedure. The details of these documents shall be readily available to all parties at the work site to make it possible for them to assist should an emergency occur. Such a list of procedures shall also be available at other appropriate locations such as the home base, operations offices, etc.

This document provides a flow chart (See **Appendix 2 HSE-HSE-FRM-022**) to follow in case of an emergency

This document is to be kept at the worksite to enable the Offshore Manager in charge to have available a form of reference and list of emergency contingency plans.

Diving Emergencies are covered by the Operations Diving Manual (**HDG-DIV-MAN-001 Section F**) and specific Project Safety Manual(s), which are issued to the site

IT IS HYDRODIVE'S MANDATORY POLICY THAT THIS FLOW CHART IS PRINTED IN COLOUR A3 PAPER, LAMINATED AND POSTED IN DIVE CONTROL AND PROJECT / OPERATIONS OFFICE.

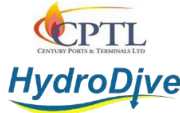
5 Decision Making Models

Emergency situations are by definition unplanned events and whilst there is some predictability in the way that certain events will unfold lends it itself to the Recognition Primed decision-making process which makes the use of flow charts and procedures extremely effective in high-stress situations, where there are complications or simultaneous unplanned events it is essential that the critical decision makers have a decision-making model from which to function, even if that is only as to which flow chart to use, or where the situation has departed from any reasonably foreseeable event and urgent decision making is required.

The following models will be employed by the three levels of command in the Hydrodiver Emergency response process

A small version of this model will appear in the margins where the model is recommended



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6 RESPONSIBILITIES

a. It is the Offshore Construction Manager's responsibility to ensure that all members of the offshore crew are fully conversant with the emergency actions.

b. **IT IS HYDRODIVE'S MANDATORY POLICY THAT A SIMULATED EMERGENCY THAT FOLLOWS THE FLOW CHART HDG-HSE-FRM-022 IS CONDUCTED AT THE START OF THE PROJECT AND AS MINIMUM EVERY 30 DAYS THEREAFTER.**

c. It shall be the responsibility of the Project Manager to issue, update and maintain this document.

To ensure that there is a coordinated response to any emergency, it is important to establish clear lines of communication. The emergency organization at the site of the incident will vary according to the location and nature of the incident but will, in all cases, be led by the OSC who will direct the onsite emergency services and be in direct contact with the EC and, if necessary, Hyperbaric Medical Centre, HydroDive Emergency Organisation includes.

For the purpose of clarity and ease responsibilities on this project should be allocated on a bronze Silver Gold model as this is an internationally recognized model and easily synchronized with other organizations

6.1.1 On Scene Coordinator (OSC) Bronze

The Offshore Construction Manager shall be the person designated as the OSC. He shall initially establish contact with the Emergency Coordinator (EC) and the Client On- **Scene Coordinator (COSC)**. **In a circumstance where the OSC is the injured person**, then the person considered most likely to be capable of responding to the emergency will be the vessel master or Diver Medic etc.

The EC will be responsible in conjunction with the vessel medic

6.1.2 Emergency Coordinator (EC) Silver

This is a member of senior management onshore. HydroDive Emergency Coordinators in project situations are usually as follows:

- Project Manager
- Designated Person Ashore
- Marine Manager

The EC will be the focal point for the entire coordination of the emergency and as such will be the initial point of contact with the OSC. The EC will determine the level of response required and will manage the response effort.

The EC will remain in contact with the COSC (after the initial contact between the OSC and COSC) at all times during the emergency, updating him on the progress of events and receiving direction as necessary. This communication allows SPDC and HydroDive senior management to manage the strategic elements of the emergency response e.g. Medevac, while EC will manage the short and medium-term response of the HydroDive personnel and assets.

A Duty Roster shall be maintained for the Emergency Coordinator. The EC who is on duty at the time of the emergency shall continue to act as the Emergency Coordinator until the emergency has been resolved.

It is also the responsibility of the EC in conjunction with the COSC to decide when to stand down the ECT. The EC is responsible for issuing a report to the HSE Manager of how the

Emergency Response was handled and lessons learnt as soon as possible after the

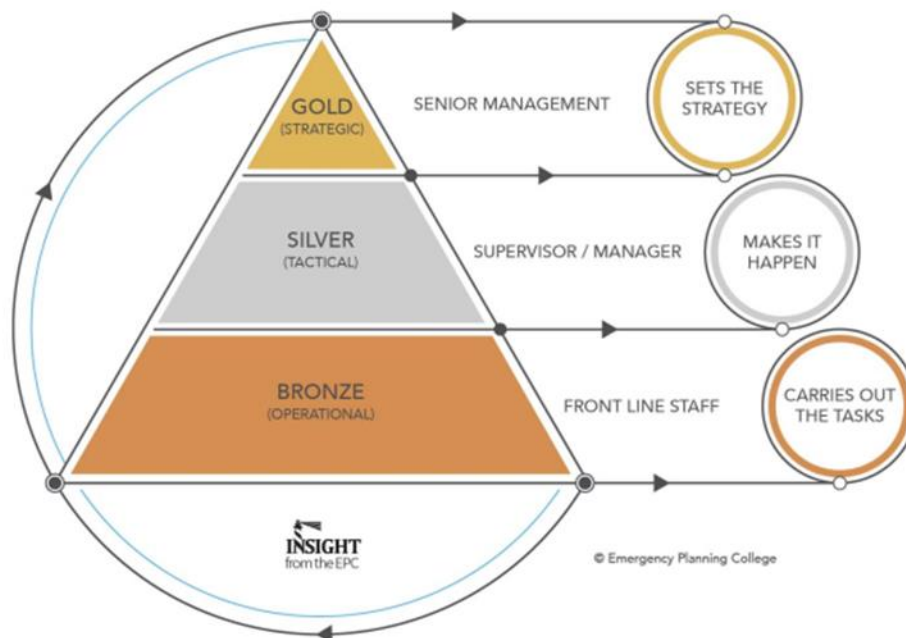
emergency.

6.1.3 General Manger – Gold

The Gold position In Hydrodive is the strategic Lead. They hold responsibility for the handling of the incident and will set an overall strategy for dealing with it. This will include any tactical do's and don'ts that need to be followed. The Gold Leader should not be drawn into making tactical decisions. Their focus stays at a strategic level however they will be responsible for reviewing and ensuring that the tactics being utilized are appropriate and sufficient to meet the strategic goals that they have set.

Some of their responsibilities will include:

- Assuming overall command and control of the incident
- Set, review and communicate their strategy
- Handling the media strategy
- Report to shareholders if appropriate
- Appoint someone to act as Chair for the Silver Team
- Consider the community and wider impact of the incident
- Define what resources or specialist skills are needed
- Appoint from The Emergency Team the Most appropriate Silver Coordinator for the Situation



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6.1.4 Emergency Coordination Members (ECM)

Emergency Coordination members include other members of HydroDive Management who are relevant in emergency situations. They are as follows:

MD/CEO	Frank Ifedi	Is the de facto GOLD Manager with overall responsibility for the incident till final close out at a corporate level
Marine and Diving Operations Director	Dave Ward	May be the delegated Gold Coordinator- or assume some delegated responsibilities or be appointed Silver controller by the GM
Personnel Logistics Manager	Latifatu Osagie	Responsible for logistics control in country and repatriation also involved with Family liaison for expatriate workers
Human Resources Manager	Modupe Oherein	Responsible for authorizing and coordination of Medical Treatment on shore
Operations Manager	Stephan Wessels	Where there are logistics and equipment concerns or the need to mobilise such will provide support to the Silver EC
Marine Manger	Steve Hardy	Senior advisor would be Silver Controller in Marine Incident and an advisor on marine capacities where needed to support another EC in incident management
Project Manger	Afam Ejidike	- Assuming overall command and control of the incident - Set, review and communicate their strategy - Handling the media strategy - Report to shareholders if appropriate - Appoint someone to act as Chair for the Silver Team - Consider the community and wider impact of the incident - Define what resources or specialist skills are needed
Legal	Tochi Nwogu	To Advise the Gold Controller on the legal ramifications of the emergency and the implications to the company and the company's legal responsibilities – enabling the Gold Controller to make the best decisions in the Company's interest

6.1.5 Emergency Coordination Team (ECT)

The ECT includes all key players in an emergency situation. It consists of the EC, OSC, ECM, HSE Manager, COSC and Hyperbaric Medical Facility (ISOS).

6.1.6 Hyperbaric Medical Centre

If the Bronze OSC requires expert hyperbaric medical assistance, he will first make contact with ISOS (HydroDive's Hyperbaric Medical Centre) and thereafter ensure that the on-call doctor liaises directly DS with the Bronze OSC always keeping abreast with developments.

6.1.7 HSE Manager

The EC will contact the HSE Manager who will be responsible for notifying and updating the Chief Executive Officer at the earliest opportunity consistent with the ultimate priority of resolution of the emergency. The HSE Manager will provide the EC with advice and any assistance the EC requires as well as maintaining a initiate the Accident & Incident and Close-out Report.

The HSE manager will be responsible for maintaining and up to date real time running log of the incident until it is decreed by the GM that the incident has been closed. These notes will assist in the close out and root cause analysis and any evaluation of the efficiency of

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the response and be used for lessons learnt going forward. These will be a legal document and constitute original notes made at the time.

6.1.8 Legal

To Advise the Gold Controller on the legal ramifications of the emergency and the implications to the company and the company's legal responsibilities – enabling the Gold Controller to make the best decisions in the Company's interest. The Legal team member will also vet all communications with the press and outside parties prior to release

6.1.9 Client-On-Scene Coordinator (COSC):

SPDC will, when circumstances dictate, appoint their own COSC who will direct the onsite emergency services e.g. medevac. He will be in direct communication with the HydroDive EC and OSC. Under normal circumstances, the Client Site Representative (CSR) is usually the COSC.

In all circumstances, it is important that there is a nominated COSC offshore at all times and that his identity is known to all personnel and other parties involved in the response to an emergency. **The COSC will be in charge of mobilizing the Duty Doctor and Aviation support from Shell**

6.1.10 On Site Medical support

Both vessels have a qualified medical doctor on board who will be able to assist in triage and provide stabilization of trauma related casualties. The goal being to maintain MEDIVAC status and transport a stabilized casualty

Both Vessels have a qualified Doctor on board as part of the project team. HydroDive personnel are trained in Basic First Aid.

Where it is a diving operation there will be at least two DMT's per shift and all diver's and supervisors will hold an in date Advanced First Aid ticket

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6.1.11 Vessel Hospital

- HD Contender is equipped with an on-board hospital/ clinic of which the primary function is a GP type service but trauma equipment is maintained in the clinic by the on board doctor

Base line equipment is as follows:

- 1. Phillip heart start defibrillator - 2
- 2. Fine test glucometer - 2
- 3. Omron electronic BP monitor - 2
- 4. Hand-held suction pump - 1
- 5. Endotracheal tubes - various sizes
- 6. Pulse oximeter - 2
- 7. Adult AMBU bag - 2
- 8. Barrier device for mouth-to-mouth breathing - 2
- 9. Mindray Beneheart D3 monitor - 1
- 10. Pneumasplints for upper and lower limbs - 4
- 11. Medical oxygen cylinder - 5
- 12. Cervical collar splints - 2
- 13. Omron Nebulisers - 1
- 14. Various stretchers - 5
- 15. Atropine
- 16. Amiodarone
- 17. Adrenalin
- 18. Adenosine
- 19. Glyceryl trinitrite spray
- 20. Magnesium Sulphate
- 21. Pentazocine
- 22. ECG Monitor
- 23. Oxygen mask of various type
- 24. Doctor grab bag for emergency- 2
- 25. Normal saline - 24
- 26. Ringers lactate - 12
- 27. 5% Dextrose water - 12
- 28. 20% Mannitol - 6
- 29. 5% Dextrose saline – 12

6.1.12 Emergency Management Off Shore

- Immediately after an incident has been identified a primary assessment shall be made as to the type(s) and severity of the incident.



- This classification of the incident will be fluid and may escalate rapidly, and the Bronze EC must be prepared to change his assessment and aware of the risks of confirmation bias and denial, in his ongoing assessment. A marine incident resulting in

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vessel incapacitation and loss of productivity may be concurrent with serious injuries to crew members.

- Thorough investigation and assessment by the first person (supervisor) on scene are essential and clear reporting.
- Immediate logging of the incident(s) should commence the bridge log will at this stage be an appropriate legal document into which detail may be recorded before a separate incident log is commenced, in this instance, the record originally made in the bridge log will for legal purposes constitute original notes, but may continue into the designated incident log- which will reference the bridge log.

The four-tiered response system is a useful decision-making tool when assessing the level of intervention required

Incident Gradings by Tier at a Company Level

TIER 00	Business as Usual	The incident has little or no impact on the ability to deliver the project. This level of incident would be covered by a first aider and would require mitigations such as a safety flash, revised risk assessment or a time out for safety.
Tier 1	Tactical Response	This is a Front Line level Response to protect People and Property and relies on the on-site team medic and DMT s This will result in the some disruption but can ordinarily be managed aby the on scene Bronze Level Controller and the project deliverable will be able to continue with minor disruption There will be a requirement foe on Shore HSE intervention and Lessons learnt implementation
Tier 2	Operational Response	This level of situation will adversely affect project delivery and require significant input from Silver Level Control. This type of situation may include mechanical incidents that prevent the company from working. Incidents that are High Potential These will be addressed by procedural review, further training Lessons learnt
Tier 3	Strategic Response	This are life threatening incidents or operational incidents that seriously threaten the vessel, project or company and require full managerial response Thes incidents will absolutely require a full mobilization of the management response team and require significant input from the Gold Level Controller

Where there is a casualty involved, the tier response on the scene is also used as a micro version of the above, these are influenced by timing and casualty management.

At the scene, the Tier system is also used

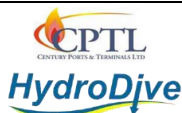
TIER	Description	Min Response Time	Assigned Party	Contact Details
0	First response by people at the emergency site	Immediate	Any of the first aid trained divers. One of the two Diver Medical Technicians (DMT) on shift.	Update names DMT on each shift. Contact by PA
1	First Aid on site	4 minutes	One of the two Diver Medical Technicians (DMT) on shift. Ships Doctor	Name Doctor Contact by PA; (room number. Internal vessel phone)
2	Patient assessment and commence treatment by site doctor / Nurse / Paramedic	1 hour	Ships Doctor Onshore medical support.	Name Doctor Contact by PA; room number. Internal vessel phone. Name and number project onshore medical support.
3	Referral to appropriate specialist hospital	Casualty Specific.	EC and COSC initiate onward medivac to Lagos	Retainer Clinic PH: Prime medical consulatans

Initial response will be determined by the incident, much of which is covered by the vessel's own process.

However, the primary actions:



- Raise Alarm
- Assess and evaluate
- Make safe the scene before attending to casualty(s)
- Assess casualty(s)
- Triage
- Where necessary move casualties to safe space
- Treat on site
- Stabilize
- Document
- The on-scene EC (Bronze) will notify the EC (Silver) as soon as possible (who will commence the on shore log and take the actions noted in the following section (Emergency Management ON SHORE)

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6.1.13 Emergency Management ON SHORE



- As soon as the duty Silver Emergency Controller is notified of an incident, based on the Bronze EC's assessment he/she will make a decision as to the level of response required. This may be escalated or deescalated as the incident develops.
- The first action will be to notify and assemble the Emergency Response Team. The designated team members will attend the Hydrodive main building in Apapa Lagos. The designated Emergency Response room is the Conference Room on the Management Floor of the main building
- This room is equipped with phone lines, white board, Internet connection, consol based speaker phones, and a large monitor screen, Video conferencing facilities are also available.
- Once the team is assembled and the initial situation understood the Gold Controller will either confirm in place the EC Silver or if there is a discipline available more appropriate to the situation will appoint that person as EC Silver for the duration of the situation
- As soon as the response room is established a running log will be maintained by the HSE lead. All decisions will be recorded in the proper format, verified by the legal representative. The GM (or his Director of operations as Gold Controller) will have the final say in the decision-making process
- It is essential that no communication leave the room without the verification of the EC Silver. All Press communications should be run through the Legal representative and verified by Gold EC.
- Where there are fatalities or serious injuries communication with the family will be via HR with the assistance of Legal.
- Hydrodive will impose a social media blackout on all failure to observe this blackout will be considered a serious disciplinary offence.



7 REFERENCES

Document Title	Document No.
Project HSE Plan	HDG-PRJ-HSE-863-03
Project Mobilization Plan	HDG-PRJ-MOB-863-01
Health, Safety & Environmental Manual	HDG-HSE-MAN-001
Accident & Incident Reporting	HDG-HSE-PR-001
Diving Operations Manual	HDG-DIV-MAN-001
Diving Emergency Flow Chart	HDG-HSE-FRM-022
IOGP 411	Jan 2021
IOGP 478	Jan 2018
IMCA D014 – International COP	Rev 3.2 May 24
IMCA D022 – Guidance for supervisors	Rev 1.3 Jun 24

Integrated Management System fleet Operation Manual	HDG-MAR-MAN-001
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8 ABBREVIATIONS AND DEFINITIONS

Term	Description
ACOE	Alternative Crude Oil Evacuation
AHT	Anchor Handling Tug
CLP	Crude Load Platform
COSC	Client On-Scene Coordinator
CSR	Client Site Representative
DDC	Diving Decompression Chamber
DFT	Domgas FYIP Terminals
DFTP	Domgas FYIP Terminals Project
DMT	Diver Medical Technician
DPA	Designated Person Ashore
DSV	Dive Support Vessel
EC	Emergency Coordinator(s)
ECM	Emergency Coordination Member
ECT	Emergency Coordination Team
ER	Emergency Response
FOT	Forcados Terminal
FRC	Fast Rescue Craft
HD	HydroDive
HSE	Health Safety and Environment
IMCA	International Marine Contractors Association
IP	Injured Person
JHA	Job Hazard Analysis
L&H	Lifting and Hoisting
MEDEVAC	Medical Evacuation
NAF	Nigerian Air Force
NCAA	Nigerian Civil Aviation Authority
OCM	Offshore Construction Manager
IOGP	International Association of Oil and Gas Producers
OSC	On Scene Coordinator
PMP	Primary Medical Procedure
PMT	Project Management Team
PTW	Permit to Work
SPDC	The Shell Petroleum Development Company of Nigeria
SPM	Single Point Mooring Buoy
TIM	Terminal Installation Manager
Hypothermia	The condition of having a body temperature approaching or below 35 degree centigrade
Hyperthermia	The condition of having a body temperature approaching or above 38 degree centigrade
Emergency	Any situation arising which requires immediate action, support to save lives and/or requires medical action and/or prevent/limit damage to property

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9 EMERGENCY ORGANIZATION & COMMUNICATION

9.1 Emergency Communications

It is essential that in establishing the initial contact and in all subsequent communications, precise details of the incident are passed using the format shown below. In the event of an Emergency, it is imperative that the **OSC (the OCM – details provided in Appendix 1) provides** every relevant detail to the **EC & DFT** to ensure a complete and appropriate response.

In all cases where any doubt exists as to the seriousness of a situation it is important that early contact is established if only to solicit a second opinion.

9.1.1 Employee Orientation Procedure

All new HydroDive employees will be required to attend an orientation program prior to employment, which will, as part of its overall content, address elements of the emergency response procedure. All project personnel shall undergo a site/vessel induction course.

9.1.2 Visitor Orientation Procedure

All visitors to the Project site will be required to go through an induction process and sign a form that they have read and understand the Visitor Orientation document. The visitor orientation will address elements of the emergency response procedure.

9.1.3 Accident Reporting

The following format is to be used in passing details, whatever mode of communication is used:

TIME AND DATE OF ACCIDENT; TYPE OF ACCIDENT:

PLACE OF ACCIDENT:

NAME(S) OF PERSONNEL INVOLVED:

WHAT IMMEDIATE ACTION HAS BEEN TAKEN INCLUDING ANY ADMINISTERING OF MEDICINES ETC:

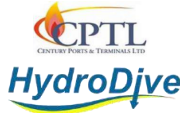
ADVISE BEST MEANS OF COMMUNICATION: I.E. RADIO, TELEPHONE (HARD LINE/SATELLITE), FACSIMILE:

ANY OTHER RELEVANT DETAILS

Incident reporting and investigation will be under **SPDC HSE MS.** (Type 2 contract)

Once **the Project Manager – (details provided in Appendix 1)** receives the initial report above it will be his sole responsibility that he and the **ECT (details of the ECT provided in Appendix 2 – Emergency Flow Chart)** complete a full investigation and close out the emergency.

- Do everything necessary to save life, and prevent damage/loss of property and environmental pollution.
- Maintaining a full record of the progress of the emergency and the response effort
- Organization of marine, sea and land transport medevac liaising with Shell PMT team
- Arranging hospitals, clinics, overseas medevac and Hyperbaric Medical cover where appropriate
- Liaison with the COSC

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- f. Answering telephone enquiries from relatives of personnel involved in the emergency.
- g. Production of list of next of kin of personnel involved at the emergency site;
- h. Liaison with Local Government services; Police, Fire Brigade, Ambulance Service etc., at the direction of the EC;
- i. Answering telephone enquiries from media, government and other external parties at the direction of the EC;
- j. Preparation of holding statements and press releases for screening by the CEO
- k. Liaison with the Chief Executive Officer as required.
- l. Complete a full investigation and close out of the emergency

9.1.4 HydroDive Interface with SPDC Organization

Close co-operation between HydroDive and SPDC Management will be essential in any emergency. This co-operation should include but not be limited to the following:

- a. Immediate and long-term operational response to the emergency, with the objective of minimizing and if possible, avoiding: -
 - Injury and loss of life to personnel.
 - Harm to the environment.
 - Damage to physical assets.
 - Damage to intangible assets such as corporate image and reputation.
- b. Investigation of the cause of the emergency
- c. Release of information concerning the cause and all other aspects of the emergency.

10 MEDEVAC & CASEVAC (Medical Evacuation & Casualty Evacuation)

Often, the terms CASEVAC and MEDEVAC are used interchangeably, but understanding the differences between these terms is crucial for effective communication and coordination during emergencies.

A “MEDEVAC” is defined as the evacuation, in emergency or not, of one or more, injured or ill personnel to a hospital onshore where further treatment will be given.

A Medevac from a worksite comprises:

- Work location to port/heliport – route confirmed and means of transport.
- Fast Boat and Onshore or helicopter transportation between site and a clinic.
- Repatriation: when necessary, managed with the assistance of the company

CASEVAC is short for casualty evacuation. This method of transport conveys injured persons to a medical facility. These evacuations are driven by urgency, so they are conducted by the closest team regardless of their medical background. A CASEVAC requires immediate action to get individuals to safety, while a medevac allows for a little more time to plan and get organized.

Generally, the most likely envisioned scenarios covered by this procedure will be MEDEVAC with the casualty stabilised by the on-vessel medical team (Medic and DMTS) prior to transport.

There may however occur situations where the risk of delaying transport outweighs the risk of the transit itself, in this instance the Vessel Master in conjunction with the Medic, if both are survivors would make the call to upgrade to a CASEVAC scenario.

The most likely scenario for this type of escalation would be a multiple casualty situation exceeding the capacity of the vessel's medical facilities.

In the event of a multiple casualty scenario, a decision can be made by the Vessel Master to initiate the vessel 'Mayday' Protocol or 'Pan Pan' this is a serious step to take and as this is not a Maritime Manual the outlines only of the decision-making protocol will be outlined here as SPDC may not be able to ensure sufficient Helicopter response in an appropriate time scale

The Master and OCM are referred to the earlier depicted decision-making model – which should be maintained as the situation escalates or deescalates

Pan Pan
Pan Pan is used to signify a situation that is urgent but not immediately life-threatening. It indicates that a vessel or aircraft requires assistance but is not in immediate peril. This might include technical issues, crew illness, or equipment failure. Using Pan Pan alerts others to stand by for further communication or assistance.
Mayday
Mayday is reserved for grave and imminent danger requiring immediate assistance. It is the most urgent distress call. Situations involving sinking, fire, or critical medical emergencies fall under this category. When Mayday is declared, it indicates that those on board are in immediate danger, and help is crucial

10.1 Depending on the severity of the incident, the appropriate medical response would be initiated.

In the event of a casualty situation occurring and evacuation being required, HD would initiate their ERP, concurrent to SPDC being informed of the immediate support requirement. SPDC would be contacted as per the contact details supplied in this plan, concurrent to the ERP being initiated by the HD OCM (OSC) on the advice of the Medic onboard / HD Contender.

Primary means of transporting an IP ashore would be by helicopter via Contender Helideck. IP will be - NCAA, Shell Aviation and Bristow prior to being used on the project. HydroDive shall ensure that the required Helicopter landing support competent persons (Helideck Team- HLO, HDA/HERTL/HERTM) are always onboard the HD Contender. For the Forcados scope the following team applies:

HLO	Helicopter Landing Officer
HDA	Helideck Assistant
HERTL	Helideck Emergency Response Team Lead
HERTM	Helideck Emergency Response Team Member

In the event of unavailability of MEDEVAC flight e.g., during bad weather or in the absence of a helicopter. The secondary Medevac means shall be utilised.

As it has been identified by SPDC that there is no designated rescue helicopter and that it may be necessary to divert an already airborne helicopter, the secondary response will always be activated simultaneously in case the Helicopter arrival time cannot be guaranteed, it is also anticipated that a helicopter that is on ground at NAF base that needs to carry a ERP team will need to transfer to Shell RA before leaving for site, this generally happens at the same time as the Chopper is being readied for movement to site, With the seats being removed. The road move has been assessed as a drive of 15 to 20 min.

Life threatening injuries Trauma Amputation Sever Blood loss Multiple casualties Cardiac Arrest Stroke Critical Illness	10 minutes from request for availability to receive confirmation that a helicopter is available – Then doctor and OCM can decide to mobilise option 2
Serious non-life-threatening Illness Psychotic episodes Mental illness	40 minutes
Illness or injury rendering un fit to work requiring on-shore medical attention	1 hour



Where it is confirmed by SPDC that a helicopter is in the air and en-route communication shall be established that will allow the target vessel to steam a course to establish an RV point that bisects the tow trajectories to minimize the return flight time as much as possible.

Where there is no confirmation of a helicopter in the air and the above time limit for aborting primary and reverting to secondary response it should be considered that a vessel underway will still shorten the response time. This is assuming that it is not a marine incident where the vessel is incapacitated

The FSPM2 Secondary MEDEVAC means is the Surfer Ose Anita 4

Contact will be made through SPDC CSR (COSC) onboard the HD Contender to transfer the IP directly to the Surfer Boat.

Mobile IP: Transfer from the HD Contender to the **Surfer Ose Anita 4** shall be via the installed HD Contender boat catcher.

Immobile IP: With the **Surfer Ose Anita 4** positioned inside the boat catcher, the Dive team will then transfer the IP using the stretcher on to the Surfer. This will be practiced as a drill on a monthly basis.

Surfer Ose Anita 4 shall then transfer the IP to the Forcados Terminal onward Medical Station via FOT Clinic, where the IP shall be stabilised. If further treatment is required after visiting the Forcados Terminal Medical Station, the IP shall be transported to Port Harcourt by the SPDC-approved helicopter as soon as possible.

Where the IP is on either the SPM and immobilized, the Omand Master will asses the weather and either use the surfer and perform a stretcher transfer or use the HD Contender's man-riding certified crane will be used to transfer stretcher to the team on the SPM, who will lift the IP to the stretcher and HD Contender certified man-riding crane used to transfer the IP to the deck of HD Contender for Medic's attention. On Medic's instruction, IP to be transferred either by the Primary or Secondary MEDEVAC means to shore.

The recovery of an Injured person from the workboat to the HD Contender shall be carried out in compliance with the approved "Workboat Diver Rescue Procedure" The Dive Supervisor onboard the workboat shall oversee coordinating this operation and will ensure all necessary parameters are in place for a safe recovery of the IP. On recovery of the IP to the workboat, the Vessels rescue boat will be deployed for transfer of the IP from the workboat to the rescue boat and then onward recovery of the rescue boat to the HD Contender for medic's attention. On medic's instruction, IP is to be transferred to the DDC and if MEDEVAC is required, either the Primary or Secondary MEDEVAC means to ashore.

The anticipated durations for the Primary MEDEVAC means. Times verified with Shell Aviation:

	TRANSIT	TIMELINE/FLIGHT DURATION
1	Bristow Helicopter notified and Helicopter prepared for emergency flight	20 minutes
2	Helicopter dispatched from NAFF BASE PH to HDC at Forcados SPM1/SPM2	50 minutes
3	IP transferred from HDC to Helicopter on Helideck	10 minutes
4	Flight from Forcados SPM1/SPM2 to Naf Base or SPDC RA	50 minutes
5	NAF BASE to Prime Clinic	10 minutes
6	Contingency (Obtaining Clearances, Helicopter Take-off and Landing, traffic)	20 minutes
Total		160 minutes



If evacuating a casualty who would be subject to flying after diving restrictions but has not been yet diagnosed with a hyperbaric illness or a diver who has a confirmed DCI. The helicopter crew will be informed via EC Silver and SPDC control and the Bronze controller will ensure that the crew on the helideck are reminded of the altitude ceiling of 1000ft. This has been verified by Shell aviation.

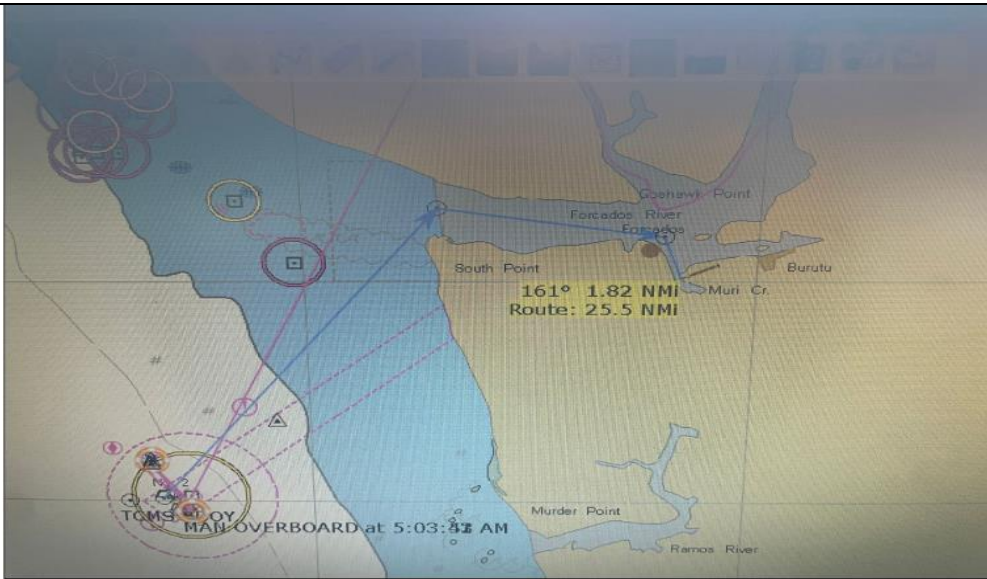
Ref: USN Rev 7 Flying after Diving

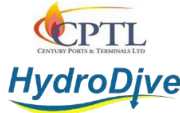
Generally anyone evacuated within 18 hours of being under pressure should be considered a residual nitrogen risk



Barotrauma IPs shall be airlifted to NAF Base PHC and retainer clinic ambulance shall convey IP to the DDC location at Atlantic Marine Yard, Port Harcourt.

Anticipated duration for the Secondary MEDEVAC means transit from site (FSPM2 & FSPM1 locations) to FOT with the **Surfer Ose Anita 4** is **75 mins** transit time. Based on a speed of 20 kts

SPEED /TIME	
22kts / 1h 09m	
21kts / 1h 13m	
20kts / 1h 16m	
19kts / 1h 20m	
18kts / 1h 25m	
17kts / 1h 30m	
16kts / 1h 35m	
15kts / 1h 42m	

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Due to the planning nature of a surfer boat this would not be achievable with a spinal cord injury casualty.

Distance from worksite to FOT is **25nM**

Emergency Drills will be conducted so as to practice and prepare as best as possible. ERP phone-call out will be conducted once HD Contender arrives in location; this is to fully ensure that:

- a) all personnel involved in ER can be contacted
- b) a suitable reaction time can be achieved.

11 Primary response plan

Helicopter from site to either Shell RA or NAF base

As soon as an airborne medivac is decided upon The EC Silver will notify the PRIME MEICAL CONSULTANTS CLINIC in Port Harcourt of the evacuation and if hyperbaric Atlantic Marine.

Prime Medical Clinic have a third party Ambulance supplier contract with Hydrodive and as soon as SPDC notify Hydrodive Silver Control of the intended arrival location and time of the helicopter, they will be notified by Silver EC and will dispatch an appropriately crewed and equipped ambulance to the designated arrival zone.

The Ambulance will be staffed by paramedics

The ambulance will transport the casualty(s) either to Atlantic Marine Yard for hyperbaric recompression or to The Prime Medical Consultants Clinic, Prime Close Rumugoba Road Port Harcourt

11.1 Capabilities of Hydrodive's On shore Emergency Clinical Support (Port Harcourt)

Hydrodive have a support arrangement with the Prime Medical Consultants Clinic In Port Harcourt. The clinic is situated less than 15 minutes by ambulance under normal traffic conditions expected in Port Harcourt from the two envisaged landing sites for a response Helicopter.

EVACUATION

The ambulances are fully equipped to transport emergencies to and from the facility or to the dedicated hyperbaric reception unit. The ambulance will be despatched with a qualified and skilled nurse with or without a doctor depending on the severity of the case being medically evacuated.

CASUALTY MANAGEMENT

The decision to use the hospital was based round a risk assessment of worst case scenario of 5 trauma casualties – this would be based on a lifting operation going wrong or an engine room incident, so considering burns smoke inhalation and trauma/ crush or amputation. Greater numbers would be a multi-agency response out with the remit of this document.

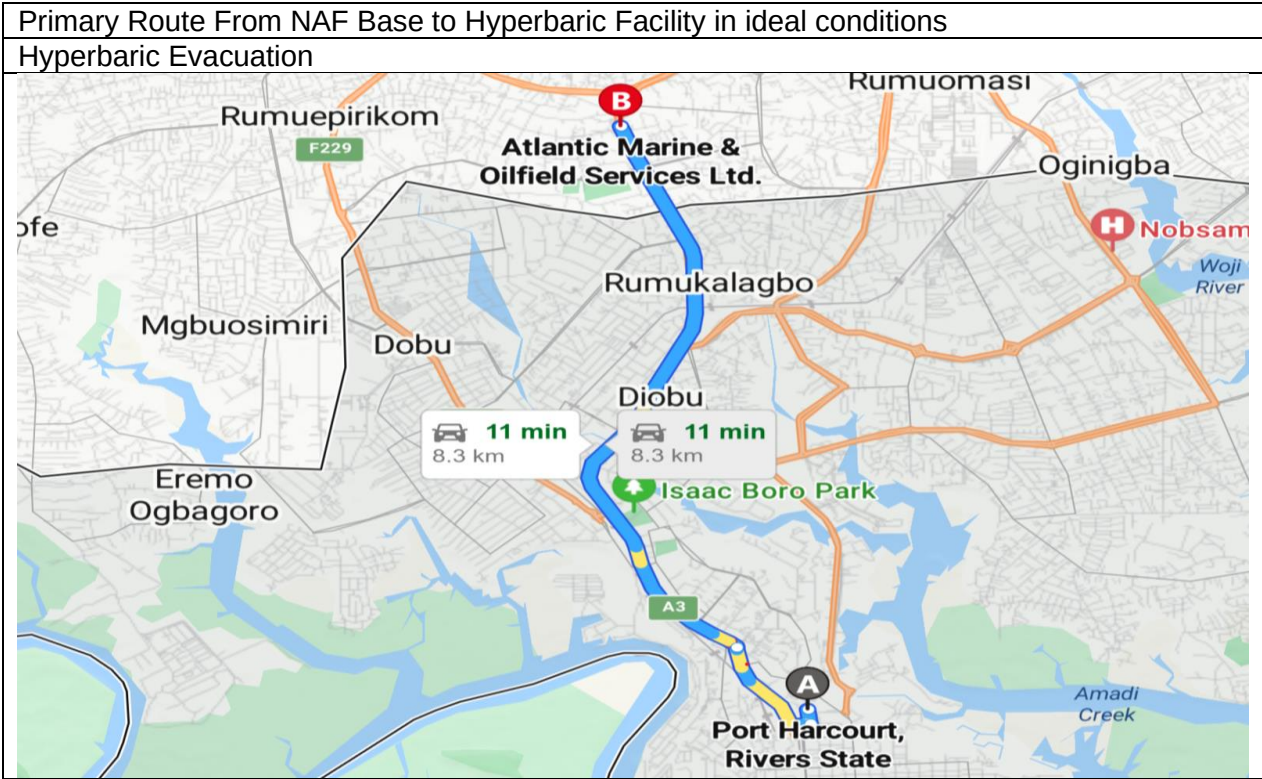
The clinic is capable of managing a worst case scenario of 5 trauma casualties with crush and or burn injuries. The emergency room has 4 beds and an emergency theatre attached to it. However, a 5th emergency in a second casualty room has one bed and currently doubles as the injection room. There is space to temporarily add a fifth bed if necessary. The fully equipped ICU has 4 beds and is manned by an internivist.

The minimum number of doctors on duty at night is 2, during the day there are a minimum of 8 doctors on duty in various sections of the hospital. 5 other doctors live in the doctors quarters

adjacent to the hospital and 2 others live about 5 minutes walk from the hospital. A minimum of 15 to 20 nurses are on duty in all sections of the hospital. There is enough manpower to handle 5 trauma emergencies at any time. On the staff are sub specialty surgeons - orthopedic, cardio thoracic, vascular, burns and plastics, neurosurgeon, urologist, ENT, maxillofacial and general surgeons. All these surgeons perform operations in the theatre that is fully equipped with a C-arm. On site there is a digital Xray and a CT scan in house with a radiographer on call 24hrs a day. There is also a 4 bed dialysis unit which is fully functional.

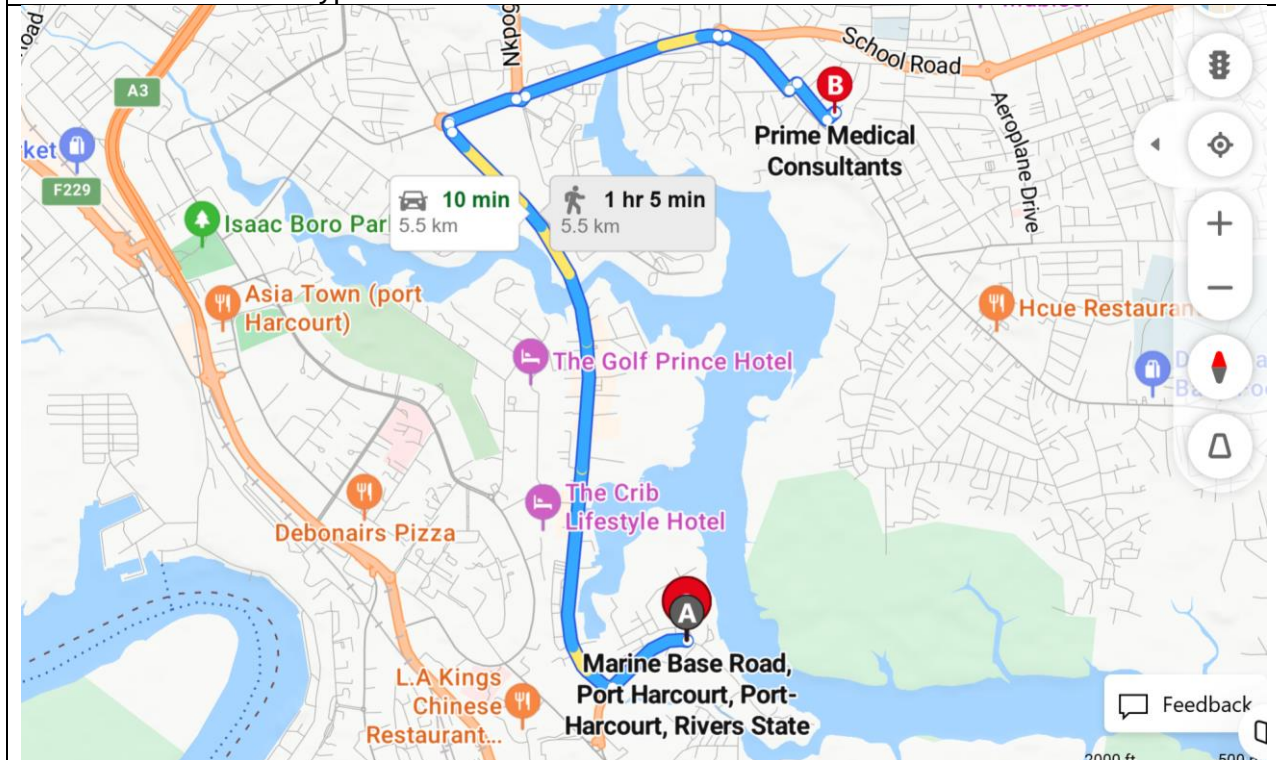
All areas of the hospital are accessible by a lift that can contain a hospital bed for ease of movement of trauma cases to various areas of the hospital (diagnostic unit, emergency room, theatre, ICU, etc)

The building is supported by
 There is a 13-man Otis elevator in Block A and a 21-man Kone elevator in Block C, to carry patients and the public to all floors of the hospital.
 There is a guaranteed uninterrupted power supply to safety critical areas such as operating theatres and ICU as a result of a dedicated transformer which ensures full current at all times,
 The clinic has three CAT generators to back up power supply and an automatic change over switch.



Primary route from NAF Base to Prime Medical Centre

Trauma or other non Hyperbaric Medical Evacuation

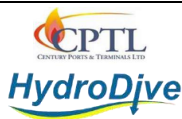


On Wednesdays, during the day, there are large street markets in the area of Atlantic Marine which cause considerable congestion. The ambulances are driven by local crews with good local knowledge. There are several alternative routes which have been driven by HD personnel. In these circumstances, the exact route will be decided by the on-the-ground crew from Prime Clinics

12 EMERGENCY DRILL PLAN

The list below indicating drills to be conducted at Project Site is not exhaustive as scenarios will determine the drills and frequency.

TYPE OF DRILL/EXERCISE	PERIODICITY
Abandon Ship Drill	Monthly and before commencement of work
Fire/Evacuation Drill	Monthly as a minimum
MEDEVAC drill	Conducted upon arrival to site
MEDEVAC drill using Stokes Stretcher	Before commencement of work
MEDEVAC drill using Stokes Stretcher to Helideck and to Surfer	Monthly
Lock-Down drill	Monthly
Unconscious Diver Recovery Drill	Before recommencement of work and then Monthly
Muster Drill	Monthly
SOPEP – Oil Spill Drill	Not required for the project
Emergency Response Team	Shall be constituted upon arrival at the site

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13 INCIDENTS OR SITUATIONS NOT IN EMERGENCY CATEGORY

All incidents, pain only decompression sickness and minor injuries where first aid is rendered shall be recorded on Accident Report Forms in accordance with the Accident/Incident Reporting Procedure HDG-HSE-PR-003.

On receipt of the report the **HSE Manager** shall copy the **General Manger**.

14 EMERGENCY SCENARIOS

Due to the rare and unpredictable nature of emergency incidents, it is not practicable to define all actions that should be followed in an emergency. However, it is important to plan and have documented, the types of actions that should be considered in a variety of emergency scenarios such as those listed in this section.

Please note that the following scenarios are covered which should not be considered as exhaustive but is rather used as a guideline on how to react to an Emergency.

1. Serious Illness, Injury or Fatality
2. Fire Onboard
3. Collision with other Vessels or Oil Field Structures
4. Grounding
5. Oil Spill
6. Man Overboard
7. Security Crime (e.g., Militancy, Kidnap, terrorist activities including banditry and piracy)
8. Missing Personnel
9. High Level of Wind and Swell
10. Diving Incidents
11. Food Poisoning (requiring Medical Emergency)

14.1 Medical Emergency Scenarios

14.1.1 Serious Illness, Injury or Fatality

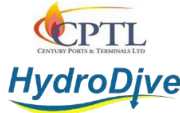
In the event an injured person is to be recovered onto HD Contender, the approved stretcher (Stokes Stretcher) will be utilized, in-order to transfer the IP to the deck of HD Contender.

If the IP is a diver with possible barotrauma, then he will be primarily treated in the HD Contender chamber by a DMT(s) under guidance from HD Hyperbaric Medical support. A DDC will always be available on the HD Contender. The Dive Medical Technician will be available at all times to operate the DDC.

In the event of the IP being unresponsive, the **EC, COSC & DFT** must be contacted as soon as possible and every attempt is to be made to continue resuscitation until the onboard Doctor takes medical charge of the injured person

14.1.2 Injured Person Requiring MEDEVAC

Following assessment by onboard Medic that an IP requires MEDEVAC, the request for the MEDEVAC should be made by the **OSC** through the **COSC & EC & DFT**.

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Primary means of transporting the injured person requiring MEDEVAC ashore would be by Helicopter via HD Contender. The IP will be transferred to the helideck via Stretcher if seriously injured.

In the event a MEDEVAC / CASEVAC is required to be carried out without a helicopter such as at night, during bad weather or in the absence of a helicopter, the IP shall receive first aid on the HD HD Contender before the available secondary MEDEVAC/ CASEVAC means is activated to transfer the IP to the Forcados Terminal Medical Station where the IP shall be stabilised. If further treatment is required, the IP shall be transported ashore via SPDC-approved helicopter as soon as possible.

The helicopter shall land at the NAFF BASE from where the IP shall be conveyed to the Shell Hospital or Prime Medical Consultants via the retainership clinic ambulance.

Refer to **Section 9** of this plan, for the MEDEVAC protocol.

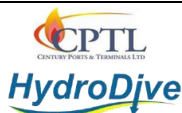
Interrupted Decompression

- If a serious incident occurs on-board HD Contender and the continuing safety of DDC operations are compromised, to the extent that it necessitates a diver undergoing decompression to be brought to the surface and to evacuate the DDC, then arrangements need to be made to transfer the diver to the nearest functioning DDC.
- The nearest functioning second DDC, is at the Atlantic Marine Yard Port Harcourt. The DDC complies with IMCA guidance. Atlantic Marine Yard shall be notified in the event of an emergency via Anthony Oguns on; 0803 552 6998 / Email aoguns@atlanticmarineoil.com or David Ekpo on 0811 121 0549 / Email: dekpo@atlanticmarineoil.com
- The MEDEVAC / CASEVAC protocol as per **Section 9** shall be activated.
- If the Primary MEDEVAC means is utilised, the helicopter shall depart for Port Harcourt NAF/ it may be necessary to reroute via Shell RA to collect Crash bag and Medical team depending on circumstance – this may add up to 1 hour to arrival time. Base, where the IP will be transferred to the awaiting retainership clinic ambulance, then conveyed to the DDC. On entering the DDC the IP is to be blown down to the pre-determined depth.
- If the Secondary MEDEVAC means is utilised, the helicopter shall depart from FOT for Port Harcourt NAF Base, where the IP will be transferred to an awaiting ambulance, and then conveyed to the DDC. On entering the DDC the IP is to be blown down to the pre-determined depth.



Fire is still the leading cause of asset loss and loss of life at sea In the event of a fire or sinking, IMO rule requires that vessels within the vicinity quickly intervene by providing rescue efforts or support.

- If a serious incident like fire such that HD Contender is unable to move, the diver under decompression will be brought to the surface put on portable O₂ and then transferred to the Ose Anita 4 at location. Estimated duration 30 minutes.
- Ose ANITA 4 will then transit to FOT where the IP will be transferred by helicopter to Port Harcourt NAF Base and conveyed by the awaiting ambulance to the DDC and be transferred into the decompression chamber.

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See below more detailed guidelines and duties for personnel involved in an emergency: -

Observer of the Incident

- Attract attention by initiating the General Alarm;
- Notify the OSC that an incident causing injury has occurred by raising alarm
- Do not attempt to move seriously injured personnel except to move them away from danger
- Apply first aid if trained to do so
- Stay with the casualty until relieved by a qualified First Aider or Vessel Medic (qualified doctor and DMTs).

On Scene Bronze Coordinator (OM)

- In the event of serious accident, ensure where possible that the scene of the incident is left undisturbed
- Collect all pertinent information

**AT THIS STAGE ACTIVATE CALL TO EC & COSC
SILVER RESPONSE**

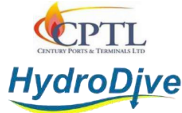
- Make the necessary arrangement for medevac to the nominated via the **EC** and **DFTP PMT**;
- Medic onboard to provide medical support as required;
- Obtain full details of injured personnel for transmission to the **EC** and to the **DFTP PMT & COSC**;
- Complete necessary Accident / Incident Report Forms and dispatch to the **HSE Manager** and **EC & DFTP PMT**;
- Determine the extent of the injury and / or symptoms. If necessary, seeking advice from the Duty Medical officer;
- In case of multiple casualties request the assistance of a First Aid Team (Doctor and DFA's) and prepare an auxiliary Sick Bay;
- Provide updated information at frequent intervals.
- On advice of the Duty Medical officer, make aware the shore medical facilities for the delivery of the injured person casualty;
- In relation to fatalities, however caused, the information required includes:
 - Names of deceased.
 - Date(s) of birth.
 - Confirmation of death by Doctor and name of Doctor or **OSC**;
 - Identity of person proving the identity of the deceased.



This is a Gold Level response. In the event of such an incident full response team will be mobilised in Lagos communication of details to family and next of kin shall only be with the permission of the Gold Controller and after absolute verification and confirmation with legal representative on the ER Team

Responsibility for Notification of the Accident

- As soon as practicable following incident containment, the Silver **EC** is to notify the **HSE Manager**
- It will be the responsibility of the **Silver EC** to arrange the notification of the Next of Kin.
- The **OSC** will provide details to the **Silver EC**.

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- d. On receipt of information the Silver **EC** will compile an immediate written briefing for the **COSC**.
- e. In the case of a fatality, the Civil Police must be informed by the Silver EC.

Accident Investigation

Preliminary accident/incident investigation should commence immediately. It is a requirement that HD shall work with the client HSE MS on Incident/Accident investigations. As much information about the accident as possible shall be gathered after the accident for example:

- i. Photographs of the site of the accident and relevant equipment should be taken.
- ii. Sketch the incident.
- iii. Obtain written witnesses' eye accounts, witnesses are to be separated when recording statements. Take witness details and have them sign their statement.
- iv. Assemble relevant documents:
 - Dive Logs
 - Dive control Tapes
 - Ships Log
 - Maintenance records
 - Written procedures
 - Narrative of events
 - Relevant equipment
 - HydroDive A&I Forms (See Accident & Incident Reporting HDG-HSE-PR-001)

14.2 Media Enquiries

- a. Any enquiries from the media directly to the Offshore Work Site should be referred to the control centre in Apapa and handled through Legal with authorization of Gold Control
- b. Be polite to the media.
- c. Do not make any statement confirming details of the accident or the reason why it occurred.
- d. If confronted by the media a typical statement, as below, to be issued from offshore.

***“At the present time the incident is being investigated by both Contractor and Company into the causes. I am unable to give you any further details until those investigations are complete.*”**

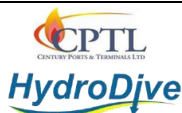
14.3 Security Emergency Scenarios

14.3.1 Serious Crime

Serious crime includes any incident warranting immediate notification to the police. The following are examples of serious crime:

- a. Significant theft, e.g. selling of HydroDives' supplies, charterers fuel etc.
- b. Deliberate Bodily Harm e.g. use of a weapon on another person, severe deliberate beatings
- c. Sabotage of HydroDive's equipment.
- d. Suspected drug abuse / possession of a controlled substance.
- e. Persistent aggression.
- f. Terrorist activity.
- g. Activities in contravention of Local Government law.
- h. Kidnapping

In these cases, the **OSC** can send persons ashore, restrain them, or take any reasonable measures to safeguard the personnel on the vessel.

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If a man is under restraint or protective custody, the **OSC** is responsible for his well-being and safety. A watch must therefore be maintained until he is put ashore or the Police/Security personnel arrive onboard.

The police are responsible for subsequent action on the crime. This action could entail for example:

- Police coming offshore to conduct investigations or to escort person(s) ashore
- Police receiving a restrained person sent ashore
- Police cautioning/searching a suspect on his arrival onshore.

If criminal actions are observed or suspected, take note of all the details and advise the OSC immediately. Guidelines for the observer of the incident and personnel with emergency duties are as follows:

- **On Scene Bronze Coordinator**

If criminal action has been observed or a complaint received:

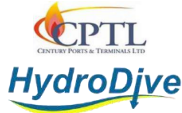
**AT THIS STAGE ACTIVATE CALL TO EC & COSC
SILVER RESPONSE**

- Privately interview complainant and any other witness
- Obtain names of suspects and any other witnesses
- In case of violence, the suspect will in most cases be under restraint. Take him into custody and detain, even if no longer violent, for his own safety.
- Arrange supervision of the suspect and escort ashore
- Any person behaving abnormally or suspected of being under the influence of drugs or alcohol is to be escorted to the OSC. Do not assume the cause, it may be due to an accident or emotional upset etc.
- Arrange for transport to shore via EC & FOT.
- Prepare a suitable room in which to detain the suspect.
- Make a stretcher or other suitable restraint.
- If appropriate go to the scene of the crime, take photographs. Rope off the area and arrange for a twenty-four hour watch to be kept. Do not remove anything until the police arrive and give notification otherwise.

14.3.2 Terrorist Activity (Piracy)

In the event of face-to-face contact with terrorists, hijackers or hostile parties the following points should be remembered:

- Keep calm and advise all personnel to do likewise
- Take appropriate measures to limit the dangers to personnel and the vessel. Initiate the HD Contender lock-down procedure.
- The aggressors probably will not understand diving operations or the operations of the vessel and may be suspicious about routine procedures. They will need to be reassured that no subterfuge is being planned.
- Offer responsible co-operation. Tension can be lowered by personnel acting normally.
- Do not return abusive treatment or aggression.
- Attempt to establish:
 - Who the aggressors are and/or what group they belong to.
 - What are their demands?
 - When they expect the demands to be met.
 - Any deadlines that have been set.
- It should be assumed that the incident will be prolonged.
- Personnel will feel isolated during an incident as they will not be aware of the steps being taken to secure their release. This may lead to antagonism against their own authorities and

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sympathy for the terrorist.

- i. The establishment of a reasonable rapport between the detainees and captors is likely to reduce the chances of violence against those being held.
- j. The establishment of a negotiating channel between aggressors and the authorities is an important step towards the successful conclusion of the incident.
- k. For divers, the following scenarios will apply in the event of piracy:

1) **If Diver is in water in free time:** Dive Supervisor to recover diver to surface immediately, On hearing the alarm, all non-essential dive personnel will leave deck area for safe zone, LARS operator, Tender, DS and Medic to recover diver then make way to safe zone . If there is no time, as a last resort, bring diver up to 3m and leave him there. Everyone clears the deck with Dive Supervisor locked in Dive Control.

The Dive Supervisor to lock the dive control from the inside using the door locking bar, blind all window with the window blinds and lay down on the floor talking to the diver and bridge. LARS made inoperable.

Diver is made aware of, and can see, the onboard gas and be prepared to use emergency air supply if required.

2) **Diver in water in deco time:** Diver will be brought to surface in a controlled manner and the appropriate omitted deco table used, Supervisor and medic to be in locked dive control to run his table with entry lock blown down. Ref No 3

3) **Diver in DDC undergoing deco or treatment:** Diver to continue staying in decompression with the correct treatment table with Chamber secured and locked as if not in use. Dive Supervisor and DMT locked in Dive control taking care of therapeutic operations keeping noise minimal. All the windows and chamber hatch locked before dive team leave for safe zone.

Entry Lock on chamber is to be left pressurized while diver in main lock undergoing deco etc.

4) **Divers on the SPM:** Immediately notification is received, divers and crew on SPM to be transferred to ERV, then to the HD Contender deck. If there is time, they join everyone to the safe zone.

If short of time ERV to pick SPM team up and follow ERV ERP concerning piracy.

- l. Broadcast a distress message, if possible, the following information is of use:
 - i. How many
 - ii. Description
 - iii. Sex
 - iv. How armed
 - v. Nationality
 - vi. Communications
 - vii. Competence
 - viii. Deployment

• **THIS INFORMATION SHOULD ONLY BE PASSED IF IT IS SAFE TO DO SO.**

In the event of a piracy attack when the diver is in water, the following actions shall be taken:

- a. Sound the piracy alarm
- b. Recover divers back to the surface
- c. If divers require chamber decompression, the Dive Supervisor shall proceed to Dive Control and lockdown
- d. Chamber Supervisor shall then proceed to the Control Panel to lockdown the Decompression Chamber
- e. All personnel to remain as quiet as possible

- f. Dive Control and Chamber Control to keep in contact via VHF radio. In the event of piracy attack when diver is undergoing decompression:
- g. The Dive Supervisor and Chamber Supervisor shall lock down the decompression chamber



The DDC's are 20-foot containers. The doors are heavy duty steel doors which are lockable from the inside.

14.4 Other Emergency Scenarios

14.4.1 Helicopter Ditching Emergency

Helicopter emergencies offshore will generally come under the following categories:

- a. Helicopter ditching in the installation area
- b. Helicopter ditching at a distance from the installation

The pilot will not land on the helideck of an installation when the helicopter is on fire but will ditch in the sea near an installation if the fire is uncontained.

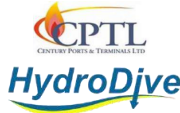
In all probability, vessels will be called to provide assistance to the ditched helicopter by the Field Superintendent. It must not be assumed however that the pilot of the ditched helicopter made radio contact with air flight control or an offshore complex or transmitted a Mayday call. Therefore, any person observing a Helicopter Ditching must immediately report the incident to the vessel Master or Officer in Charge.

On Scene Coordinator

- a. Obtain all the facts from the incident observer(s);

**AT THIS STAGE ACTIVATE CALL TO EC & COSC
SILVER RESPONSE**

- b. Proceed to the scene of the incident maintaining radio watch and listening for a call from the **EC**, **DFT** or **COSC**;
- c. Suspend all radio, telephone and fax communication except for those dealing with the emergency;
- d. On reaching close to the scene of the incident, unless directed otherwise by the **COSC** or **Master** of the Vessel commence operations to save life;
- e. Launch rescue boat;
- f. Seek assistance by directing other vessels to the rescue scene;
- g. Prepare medical reception facilities onboard, stretchers, medicines etc.;
- h. Provide updated information to the **EC** & **DFTP** at frequent intervals.
- i. If personnel are rescued obtain full details as follows:
 - i. Names of rescued
 - ii. How many passengers were on-board?
 - iii. Did they see any other survivors unaccounted for
 - iv. Did they see any injured or fatalities?
- j. Advise on medical support required and nature of injuries of survivors.

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14.4.2 Fire Onboard Vessel

Fire onboard a vessel is probably the most life threatening of any scenario that can be encountered.

It is the most common shipboard emergency and results usually due to human carelessness, combined with the many fire hazards found onboard, such as paint lockers, fuel tanks and lines, galley grease, smoking in bed, faulty electrical equipment and inadequate stowage space. It is the most terrifying because although any fire is frightening, shipboard fires may have to be fought in confined, unlit spaces below decks, plus the potential heat of a shipboard materials fire, is very high.

The only way to deal with this situation is by prompt action from a well-trained fire-fighting team and well-trained back up support. Any delay in the assembly of the fire-fighting team and support team can lead to the fire quickly becoming uncontrollable leading to the death and serious injury of other members of the crew and passengers.

The Fire Triangle

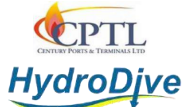
- a. All HydroDive and third-party personnel should fully understand the fire triangle. It takes the following to make a fire:
 - i. A combustible material or fuel (Paper, Bedding, Galley Grease, Fuel, Oil, Paint, Thinner, Electric Current)
 - ii. Heat (Welding, Cigarettes, Spontaneous Combustion, Galley Stove, Electricity, Hot Machinery)
 - iii. Oxygen (Surrounding Air, Fan Forced Air, Open Doors)
- b. Eliminate anyone of these three things and you extinguish the fire.
- c. Heat is eliminated by applying water which cools the fire and reduces the temperature of the burning material to below the combustible temperature of flash point.
- d. Oxygen can be eliminated by applying carbon dioxide (CO₂) or dry chemical (DC) and restricting oxygen from the area around the burning material, or by applying foam which blankets the burning material.
- e. Eliminating fuel is usually a manual action, such as throwing the burning substance overboard or shutting off a fuel oil valve, etc.
- f. Fire can occur at any of the following locations:
 - i. On the deck
 - ii. In the accommodation or galley areas
 - iii. In the store rooms and workshops
 - iv. In the engine or machinery room

Observer of The Outbreak of Fire

- a. Immediately fire is suspected either as a result of smell or detection by sight, sound the alarm by breaking the glass cover of the fire alarm and pressing the switch
- b. Report immediately to the wheelhouse.

On Scene Coordinator

- a. The Master will assume **OSC** position and is in overall charge;
- b. Master will ensure the fire alarms are still sounding and if near other vessels sound the vessels horn to attract their attention with one long blast followed by other long blasts on the whistle;
- c. Master will mobilize fire squads as per the Muster List and ensure that emergency operations are underway.

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- d. Make public address over the P.A. system alerting everyone to the situation onboard. If the vessel is in port the local Fire Authority should be informed of the nature of the emergency;

**AT THIS STAGE ACTIVATE CALL TO EC & COSC
SILVER RESPONSE**

- e. Check muster list to ensure that all personnel are out of the accommodation and accounted for;
- f. If all personnel are not accounted for, action must be taken to identify the last known location of missing person(s) and fire-fighting team supported by personnel equipped with, and trained in the use of, breathing apparatus should, without endangering themselves or others, try to locate and recover the missing personnel;
- g. Should the fire-fighting operations, including the use of engine room smothering system, prove inadequate to fight the fire without the possible loss of crew members or passengers lives, do not take chances and prepare to abandon ship;
- h. If situation deteriorates and the vessel has to be abandoned the Master will sound seven short blasts followed by one long blast on the whistle to muster all personnel to the life raft stations;
- i. Master will make MAYDAY transmission on ships radio and utilize ships distress pyrotechnics;
- j. Abandon vessel in orderly manner ensuring that pyrotechnics, hand-held radios, life raft radio and EPIRB are taken onboard the life rafts;
- k. Move life rafts clear of the vessel and collect all life rafts together and secure one to another.
- l. Continue sending distress signals and await rescue.
- m. If a diver is undergoing decompression during a fire outbreak, details of the emergency response activities are covered under Interrupted Decompression in Page 15. Immediately, a fire outbreak requiring abandonment, the dive supervisor will initial emergency decompression to surface while the DMT proceeds to put the diver on oxygen.

14.4.3 Collision with Other Vessel or Oilfield Structure

A Collision incident may result in one or more of four types of misfortune:

- a. The vessel may be damaged and taking on water, the seriousness of which must be immediately determined and controlled;
- b. Personnel injury may occur and need to be attended to;
- c. Fire is a possibility;
- d. If damaged badly, it could become necessary to abandon ship.

In the event of collision with another vessel or oil field structure, prompt action should be taken to determine the damage to the vessel and consequential effects.

If a diver undergoing decompression has to be evacuated following vessel collision with another structure, the dive supervisor will immediately blow down the diver and immediately be put on oxygen and transferred to Ose Anita 4 to safety.

On Scene Coordinator

- a. Has injury been sustained by any crew member or passenger onboard the vessel?
- b. Is the stability of the vessel in jeopardy as a result of flooding of compartments and is the vessel in danger of sinking? Check all tank soundings for flooding / leakage;
- c. Has fire started in any compartment as a result of the collision?
- d. Is the vessel in danger of sinking or is there danger to life?
- e. Alert SPDC Clients Representative.

AT THIS STAGE ACTIVATE CALL TO EC & COSC SILVER RESPONSE

- f. In the case of (a) follow directions as per Paragraph 1 Serious Illness, Injury or fatality. Clarify the safety of the vessel with regard to stability and take such action as is required to ensure adequate safe stability. If safe stability can't be attained prepare to abandon ship and sound the distress signal of seven short blasts followed by one long blast on the whistle.

Duties of The Master in Cases of Collision

- a. In the case of collision with an oil installation or collision with a vessel it is the responsibility of the Master or person in charge of the vessel, if insofar as he can do so without danger to his own vessel or crew and passengers (if any):
- To render to the other vessel, her Master, crew and passengers (if any) such assistance as may be practicable, and may be necessary to save them from any danger caused by the collision, and to standby the other vessel until he has ascertained that she has no need of further assistance, and also;
 - To give to the Master or person in charge of the other vessel the name of his own vessel and the port to which she belongs, also the names of the ports from which she comes and to which she is going.
- b. In case of collision with a vessel or platform it is the responsibility of the Master or person in charge of the vessel, if in so far as he can do so without danger to his own vessel or crew and passengers:
- If the platform is unmanned, try to determine the extent of damage to the platform by visual means and try to determine by visual means if there is any leakage of oil / gas with the possibility of resulting pollution or explosion consequences.
 - After a collision with an oil field structure (manned or unmanned), move the vessel to safety (up wind side of the platform) as quickly as possible. This will move the vessel out of the way of oil on the water or gas in the air.
- c. On completion of rendering assistance (if required) and making sure the vessel is seaworthy; proceed to the nearest safe port.

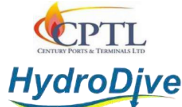
14.4.4 Grounding

In the event of grounding the following action should be taken:

On Scene Coordinator

AT THIS STAGE ACTIVATE CALL TO EC & COSC SILVER RESPONSE

- If in an oil field alert the SPDC Marine Coordinator.
- Master will stop all main propulsion;
- Check to see if injuries have occurred to crew and passengers as a result of grounding;
- Master will take the soundings of all tanks to ascertain if the hull integrity has been breached;
- Check the state of the tide to ascertain if flooding or ebbing;
- If the tide is flooding, discharge ballast/fresh water and by using the main engines in reverse attempt to move the vessel off bottom;
- If the vessel is unable to move under its own power the **EC** will, via communication with SPDC Marine Manager, arrange for a towing vessel to be made available at the time of the next flood tide;

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- h. While waiting for the tow off, the Chief Engineer should ensure that he uses only the sea suction inlet furthest away from the point of grounding, to ensure that no sand etc., is drawn into any of the auxiliary engine cooling systems;
- i. Prior to commencing the tow off, communication should be established with the towing vessel.
- j. Discuss in detail the plan and the sequence of events with the Master of the towing vessel;
- k. At the time of commencing the tow off, ensure that no personnel are on the open back deck of the towing vessel;
- l. Once the vessel is safely afloat, under its own power and the towing hawser is disconnected, Master will check once again the soundings of all compartments to ensure no ingress of water;
- m. Chief Engineer will check all water filters on all engines and flush all systems.

14.4.5 Oil Spill

All offshore bulk cargo transfers should be carried out in compliance with the vessel "Procedure for Safe Offshore Bulk Cargo Transfer". Transfers are the responsibility of the vessel Master and crew.

- a. Oil spills that do occur during offshore operations usually occur at the time of fuel oil transfer during bunkering operations and may result from an overflow of tanks or a burst hose. The total amount of oil spill will depend on the time taken to stop the operation of the transfer.
- b. Other possibilities could be as a result of collision or grounding of the vessel.
- c. The type of oil involved will be marine-grade diesel oil.
- d. In the event of an oil spill during bunkering operations:
 - i. Stop the transfer
 - ii. Drain back the hose content if possible, close all valves
 - iii. If the tank has been overflowing should be drained back into an empty tank or tank with less ullage using an internal line system
 - iv. Check all scuppers are closed and watertight to prevent as much oil as possible from flowing into the sea.
 - v. Maintain fire watch - rig up fire hoses, prepare fire-extinguishers
 - vi. Assess the amount of diesel oil spilled into the sea.
 - vii. Inform SPDC Marine Coordinator with regard to the extent of pollution
 - viii. Inform the Port Authorities if the incident happens within port limits.
 - ix. Inform HydroDive **EC & DFTP**.
 - x. Commence clean-up of the residual oil from the decks of the vessel using the spill kit
 - xi. Complete all relevant reports and forward as soon as possible to HSE Manager

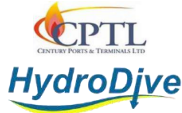
14.4.6 Man Overboard

The primary and most obvious action, in the event of a man falling or being swept overboard, is to retrieve him from the water rapidly before he drowns or becomes a victim of hypothermia or hyperthermia.

Instantaneous actions can save a life. Indecisiveness can cost a life.

The immediate necessity is to get something that floats (preferably a ring buoy c/w illumination attached) into the water close enough so that the man can swim to it and hang on (until a zodiac/FRC is sent to him). Of next importance is to get a Zodiac/FRC as quickly as possible and get him out of the water before hypothermia/hyperthermia paralyses him, exhaustion overcomes him, or wave action drowns him, even while he is hanging on to a ring buoy. To accomplish this, several instantaneous actions must be performed by the Vessel Master or the Dive Supervisor. It must be noted that all crew at work will be donning type 1 PFD.

On sighting a man fall overboard or on sighting a man in the water, the observer must shout "Man Overboard", indicating the side, port or starboard.

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The officer on duty should carry out the following procedures:

- a. Helm over towards the side the man is in the water;
- b. Release the man overboard life ring and combined light and smoke signal;
- c. Sound the man overboard signal (on ships whistle - three prolonged blasts) and raise the general alarm (flag "O" to be raised if appropriate);
- d. Watchman to be charged with keeping the man in sight at all times;
- e. Execute the man overboard manoeuvre;
- f. Transmit message to other vessels in the area for assistance if required;
- g. Cancel message and send an all clear message to other vessels when man is recovered onboard

**AT THIS STAGE ACTIVATE CALL TO EC& COSC
SILVER RESPONSE**

- h. If the man is lost from sight;
- i. Alert the crew and prepare the required emergency equipment;
- j. Alert field Marine Coordinator;
- k. Complete all relevant forms and forward as soon as possible to the **EC & DFTP**.

Man-Overboard Maneuver

- a. Send an MOB Boat (small boat or zodiac) to the man for rescue.
- b. **Illuminate** - If at night, a search light should be manned immediately, and the light shone on the man in the water continuously during vessel maneuvering.
- c. Furthermore, as a standard procedure, when surface visibility is severely restricted due to darkness at night or fog it is required that crew wear a life jacket and being clipped on to a well-secured jack line especially while working close to the edge on the deck for any reason. A secured grabline or belt to the life jacket will protect MOB due to deck exposed to night moisture or an unexpected steep wave.
- d. If after proceeding back down the original track line to the estimated location of the man in the water and he is not immediately located then, launch the rescue boat to search the immediate area.
- e. **Retrieve the man.** Men on deck should be standing by in life jackets to pull the man out of the water quickly. This should be accomplished by heaving him a line and pulling him to a low freeboard location if possible. Someone in a lifeboat should search the immediate area.
- f. The vessel should then begin an expanding square search pattern starting at the location of the ring buoys. The square is begun with two 5-minute legs and the square expands outward by increasing the legs in 5-minute increments.

EXAMPLES OF DISTRESS TRANSMITTING PROCEDURES

The VHF radio calling channel for use within the operational areas can be found in the individual Marine Standing Orders provided by the Operating Companies.

All vessels should have two VHF radios. One VHF radio should be assigned to this channel at all times. The remaining VHF radio should be assigned to Channel 16.

To transmit a distress message using VHF radio, follow the following procedures:

- a. MAYDAY MAYDAY MAYDAY (Spoken three times)
- b. THIS IS (NAME OF VESSEL). (Spoken three times)
- c. MY POSITION IS (.....) (e.g. Three miles east of XYZ)

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- d. SEND NATURE OF DISTRESS (e.g. I am on fire and require immediate assistance).
- e. Repeat message if there is no response from other vessels.

On hearing the distress message, all VHF radio communication of other vessels should stop immediately and wait for the message to be transmitted. Vessels which have received the distress message should wait before replying unless their position is close by. Vessels on standby duty should not leave their location unless instructed to do so by the SPDC Marine Coordinator.

- a. Call up the distress vessel
- b. Give details of your vessel's position and ETA and the type of assistance you can provide
- c. Proceed at full speed to the location of the distressed vessel
- d. Alert the crew and prepare the required emergency equipment - fire hoses, BA sets, rescue boat, medical reception and medical facilities etc.

Emergency Coordinator (Silver Level EC) – A Senior Management onshore. The **Silver EC** will be the focal point for the entire coordination of the emergency and as such will be the initial point of contact with the **OSC**. The **EC** will determine the level of response required and will manage the response effort.

A **Duty Roster** shall be maintained for the Emergency Coordinator Silver.

Silver EC will remain in contact with the **COSC** (after the initial contact between the **OSC** and **COSC**) at all times during the emergency, updating him and the SPDC PMT Team on the progress of events and receiving direction as necessary. This communication allows SPDC and HydroDive senior management to manage the strategic elements of the emergency response e.g. Medevac, while **EC** will manage the short and medium term response of the HydroDive personnel and assets.

A **Duty Roster** shall be maintained for the Emergency Coordinator Silver.

The **Silver EC** will provide advice and direction as required by the emergency, this advice will not relieve the Project Manager / Senior Supervisor (Superintendent) from their obligations or responsibilities for the safety of their personnel under his command at all times. The **Silver EC** shall continue to act as the Emergency Coordinator until the emergency has been resolved.

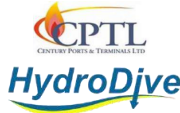
It is also the responsibility of the **Silver EC** in conjunction with the **COSC** to decide when to stand down the **ECT**. The **Silver EC** is responsible for issuing a report to the HSE Manager of how the Emergency Response was handled and lessons learnt as soon as possible after the emergency.

Emergency Coordination Member (ECM) - The **EC** will appoint, as necessary, members of HydroDive management to assist him during the emergency.

Emergency Coordination Team (ECT) – Consists of the **EC, OSC, ECM, HSE Manager, COSC and Hyperbaric Medical Facility TAC HEALTH CARE**

Hyperbaric Medical Centre – If the **EC** requires expert hyperbaric medical assistance he will first make contact with **TAC HEALTH CARE** (See Appendix 11 for details) and thereafter ensure that the on-call doctor liaises directly with the **OSC** but always keeping the **EC** copied.

HSE Manager - The **EC** will contact the **HSE Manager** who will be responsible for notifying and updating the **Chief Executive Officer** at the earliest opportunity consistent with the ultimate priority of resolution of the emergency. The HSE Manager will provide the **EC** with advice and any assistance the **EC** requires as well as initiate the Accident & Incident and Close Out reports

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(see **Accident & Incident Reporting HDG- HSE-PR-001**)

Client On-Scene Coordinator (COSC): SPDC will, when circumstances dictate, appoint their own COSC who will direct the onsite emergency services e.g. medevac and who will be in direct communication with the HydroDive's EC.



Client Site Representative (CSR) is the designated COSC and HydroDive will at no time and under any circumstances appoints anyone as the COSC.



SPDC may choose to designate another person as COSC.

In all circumstances it is important that there is a nominated **COSC** offshore at all times and that his identity is known to all personnel and other parties involved in the response to an emergency.

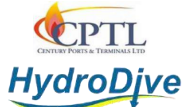
Dedicated Medic: A qualified medical doctor (Medic) will be stationed on-board HD Contender to cover all personnel and vessels on site and to support in all emergency medical scenarios at the worksite. Approved IMCA drugs and First Aid materials will be made available by HydroDive to cover the POB.

HD Contender:

Medic – Dr Chibuike Nwosu/Dr Newton Atenaga (B2B); Vessel Ext 8057

Captain - +234 705 273 5802

3 Dedicated First Aiders will always be available on vessels. HydroDive personnel are trained on Basic First Aid. 4 x DMTs (2 x DMTs per shift) shall be made available on the crew mobilisation to site.

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APPENDIX 1: CONTACT DETAILS

Below are detailed the primary contact numbers for vessel and key personnel.

Names and contact details for all Emergency Response team members are identified and are to be found in Appendix 2.

HYDRODIVE

OFFSHORE		
Offshore Manager (OSC) BRONZE	Name:	Dean Golding / Jake Ward
	Telephone	+44 (0)7943006825 / +44 (0)7964226431 / +234 915 754 1369 / +27 82 909 0079
	HDC Email:	offshoremanager.863.01.24@hnlprojects.com
HD Contender	Call Sign:	5NUU3
	Telephone:	+1 281 824 1970
	Email:	captain@hdcontender.com
	Email:	bridge@hdcontender.com
	Radio:	VHF Ch 12/16

HYDRODIVE CONTACTS		
ONSHORE		
Hyperbaric Medical Support - ISOS	Telephone:	+44 (0) 20 8762 8347 1rclonops@internationalsos.com
Project Manager (EC)	Name	Afam Ejidike
	Telephone	+234 (0) 817 381 4385, +234 (0) 915 532 6366
	Email	a.ejidike@hydrodive.com
HSE Manager	Name	Kene Anyabolu
	Telephone	+234 (0) 8035250009
	Email	k.anyabolu@hydrodive.com
Emergency Coordinator (DPA)	Name	Bolaji Olarewaju
	Telephone	+234 (0) 8039753310
	Email	vessels@hydrodive.com

Emergency Coordinator 2 (Marine)	Name	Steve Hardy
	Telephone	+234 (0) 8138612713/(0)7032580471
	Email	s.hardy@hydrodive.com
Secondary DDC	Name:	Anthony Oguns
Atlantic Marine Yard Port Harcourt. 114 Aba Rd. PHC	Telephone:	+234 0803 552 6998
	Email:	aoguns@atlanticmarineoil.com
Retainer Clinic Bonny: ST. Peters Clinic., 19 New Layout, Bonny.	Telephone	+234 (0) 8065857149 or +234 (0) 8051612747 or +234 (0) 7054988850
Retainer Clinic Lagos: Lagoon Hospital, 8 Marine Road, Apapa, Lagos.	Telephone:	+234 (0) 1-8707011 or +234 (0)1-6283813 - 5 or +234 (0) 808718137
Retainer Clinic Warri: Lily Hospital Ltd., 6 Brisibe Lane, Warri	Telephone:	+234 (0) 7053352417 or +234 (0) 7053351743 or +234 (0) 7053351700 - 49
Retainer Clinic Port Harcourt: Prime Medical Consultants, 8 Prime Close Rumuogba, Port Harcourt.	Telephone:	+234 (0) 8033134822 or +234 (0) 8022237553 or +234 (0) 8034538188.

SHELL ONSHORE

SPDC BASE		
PROJECT MANAGER	Name:	Chris Ubane
	Telephone:	08070242321/08070342321
	Email:	chris.o.ubane@shell.com
PROJECT OWNER	Name:	Mobolaji Popoola
	Telephone:	08070590918
	Email:	mobolaji.popoola@shell.com
S&E LEAD	Name:	Etsano Emmanuel
	Telephone:	08070341725
	Email:	Emmanuel.etsano@shell.com
PROJECT LEAD (First Contact in Emergency)	Name:	Kent Wolu
	Telephone:	08070311450
	Email:	k.wolu@shell.com
PROJECT ENGINEER	Name:	Kent Wolu
	Telephone:	08070311450
	Email:	k.wolu@shell.com
ASME (Diving)	Name:	David Mooney
	Telephone:	+234 816 408 1842
	Email:	d.mooney@shell.com

SPDC BASE

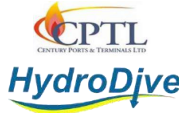
SECURITY LEAD, DFT PROJECT	Name:	Hesketh David
	Telephone:	0807 031 0837
	Email:	david.hesketh@shell.com

SPDC OFFSHORE

CLIENTDIVING REP 1.	Name:	Andrew Routley
	Telephone:	+2348077855335
	Email:	andrewroutley@hotmail.com
CLIENTDIVINGREP 2.	Name:	Peter Uys
	Telephone:	+234 9151761586
	Email:	uyu.ptu@gmail.com
Client Site Rep	Name:	
	Telephone:	
	Email:	

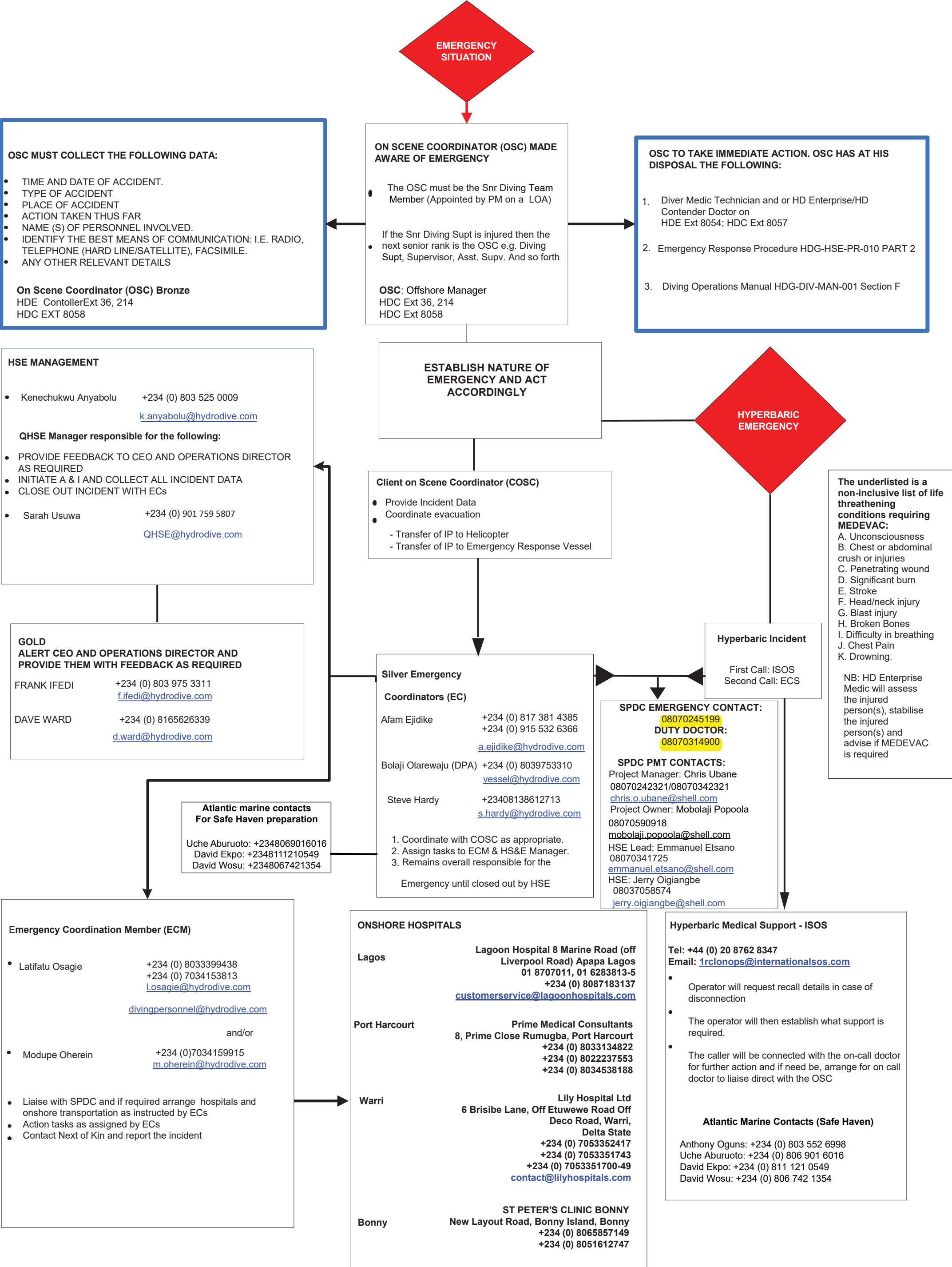
SHELL FOT

Position	Name	Telephone	E-mail
Manager, Security, Major Project & Wells	Ogburn Johnson	08070339408	Johnson.ogburn@shell.com
Project Security Coordinator	Omole Rotimi	08096902441	Rotimi.omole@shell.com
Security Lead, DFT Project	David Hesketh	08070310837	David.hesketh@shell.com
DSS – FOT	Jonathan	08057326740	
CLP Control room	N/A	57381	N/A
Forcados TIM	Acholonu, Anny	45004/31369	Anny.Acholonu@Shell.com
Forcados control room		45112/31374	
SPDC Security Control Room		45095 0807022292	
Shell Contact warri Airstrip	Utieyintsola, Ayo	48436	Ayo.utieyintsola@shell.com
SPDC Medical	Medical Emergency	+2348070345152 +2348070245199	
SPDC Fire Service	FOT Emergency No.	199 +2348070245199	
SPDC Oil Response	FOT Emergency No.	199 +2348070245199	
OHN/Duty Doctor	Medical Emergency	32032/28518 & +234 80702 28518	

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APPENDIX 2: FSPM1 & FSPM2 SITE SPECIFIC EMERGENCY FLOW CHART

1 Page to Follow



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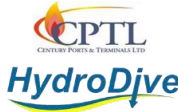
APPENDIX 3: TIERS OF MEDICAL EMERGENCY RESPONSE

1 Page to Follow

	Description	Min Response Time	Assigned Party	Contact Details
0	First response by people at the emergency site	Immediate	Any of the first aid trained divers. One of the two Diver Medical Technicians (DMT) on shift.	Update names DMT on each shift. Contact by PA
1	First Aid on site	4 minutes	One of the two Diver Medical Technicians (DMT) on shift. Ships Doctor	Name Doctor Contact by PA; (room number. Internal vessel phone)
2	Patient assessment and commence treatment by site doctor / Nurse / Paramedic	1 hour	Ships Doctor Onshore medical support.	Name Doctor Contact by PA; room number. Internal vessel phone. Name and number project onshore medical support.
4	Referral to appropriate specialist hospital	Casualty Specific.	EC and COSC initiate onward medivac to Lagos	Retainer Clinic Lagos: Lagoon Hospital, 8 Marine Road, Apapa, Lagos.

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APPENDIX 4: DIVING EMERGENCY

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15 DIVING EMERGENCY EVENTS AND MITIGATIONS

The following emergency response situations attempts to identify emergency scenarios that may occur during the deployment of an air Diver. As not all site conditions and emergencies can be covered by generic procedures, the Senior Diving Supervisor must use his discretion and resources available to him in dealing with any emergency that may arise.

15.1 Fire in Dive Control (Diver in Water)

Supervisor Actions	• Activate general alarm and • Inform Bridge, DDC operator, Divers and Dive Techs. • Instruct divers to return to Dive baskets • Put on emergency breathing apparatus and comms • Activate fire suppressor system and/or put out fight fire with appropriate fire extinguisher. • Ensure oxygen supply to Dive Control is isolated on deck. • When the requirement becomes obvious, don BA sets and establish communications with the diver for recovery • If loss of comms, use surface standby diver to relay message or flash hat light signals • Recover Divers to surface • Secure gases at quad and vent lines
Diver Actions	• Return to Dive Basket and prepare to leave bottom • Wait for supervisor instruction for recovery by LARS to surface

15.2.1 Fire in Vicinity of Chambers

Surface Actions	• Activate general alarm and • Inform Bridge, DDC operator and Dive Techs • If there is a secure escape route and are confident of being able to tackle the fire without exposing yourself to danger, then do so using the appropriate fire extinguisher. • Liaise with the vessels fire party. • If Divers in DDC, Inform Divers to prepare for possible evacuation. • Monitor chambers for any depth increase. • Secure Oxygen and electrical supplies to chamber area. • Open deluge system
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15.2 Loss of Communication to The Diver

Failure of communication can result from a number of life threatening and non-life-threatening events. In the interests of Diver safety, the Diving Supervisor must assume that a loss of communications is, or may become, a life-threatening event and react accordingly.

Communication failures may arise from a number of factors, including but not limited to: -

- Discharged batteries in the Divers Radio or loss of main power
- Loose contacts in the Divers Radio
- Loose fitting jacks or Banana plugs
- Water ingress into the communications cable or splice
- Failure of Divers hat microphone
- Failure in Divers hat connection to internal speakers and microphone
- Severance of communications cable
- Total severance of Divers Umbilical
- Unconscious Diver

Surface Indications	• No verbal response from diver • No sound of diver breathing
Subsea Indications	• No audio from supervisor

15.2.1 Loss of Communications (Diver Responding)

Supervisors Actions:	• IMMEDIATELY STOP all moving parts (crane / winch wires and equipment moves etc.) • Immediately switch to a secondary air supply. • Advise the Diver that communications have been lost (communications failure may be one way only) • Ask Diver to signal Tender with pulls on the umbilical. • Flash Divers hat light four times to alert Diver of communication failure. • If the working Diver does not respond to voice communications or umbilical line pulls, the Diving Supervisor to follow Loss of Communications and No Diver Response procedure. • If the working Diver responds to voice communications or umbilical pulls, instruct diver to return to basket • Recover diver to surface
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Tenders Actions:	- Give the standard signal of one (1) pull on the umbilical. If the working Diver is OK, his response will be one (1) pull, "I am all right". - When advised by the Diving Supervisor, Tender will give the Diver four (4) pulls on the umbilical to indicate to the Diver to leave bottom. The Diver will respond with four (4) pulls to indicate that he is leaving bottom.
Divers Actions:	- Immediately the Diver is aware of lost communications he is to return to dive basket and prepare to leave bottom. - On receiving one (1) pull on his umbilical, Diver is to respond with one (1) pull and advise topside of such (failure of topside to reply will indicate problem with communications) - Clear umbilical from any obstructions - Give four (4) pulls on the umbilical to indicate if you are ready to leave bottom.

15.2.2 Loss of Communications (Diver Not Responding)

For loss of communication with no diver response, follow recovery of unconscious diver procedure.

15.3 Loss of Breathing Supply to Diver

Failure of normal air supply can result from a number of life-threatening and non-life-threatening events. Loss of normal air supply may arise from a number of factors: -

- L.P. Compressor cut out and receiver bled down.
- Rupture of surface hose from L.P. Compressor
- Failure of Regulator at H.P. Quad
- Rupture of surface hose from H.P. Quad
- Crushed Hose
- Failure of umbilical fitting
- Rupture of air line in umbilical
- Total severance of Divers umbilical

Throughout the dive the Diving Supervisor remains in communications with the Diver and can monitor the breathing activity of the working Diver.

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Surface Indications	• Loss of pressure on diving panel • Diver reports gas tight • Diver will report switching to emergency gas supply (bailout) • Noise of escaping gas
Subsea Indications	• Diver gas becomes tight/stops • Diver switching to emergency gas supply (bailout)

15.3.1 Loss of LP Compressor / H.P. Quad pressure

Supervisors Actions:	• On noticing drop in primary air pressure on dive panel gauge, switch to secondary air supply • Rectify loss of primary air supply by checking L.P. compressor and/or charging H.P. Bank • Assess need for Diver recovery • Recover diver if necessary
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15.3.2 Failure or Rupture in Air Supply Delivery to Panel

Supervisors Actions:	• On noticing drop in primary air pressure on dive panel gauge, switch to secondary air supply • Recover Diver to surface • Rectify loss of primary air supply: Replace hoses or fittings and/or Replace regulator
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15.3.3 Failure of Umbilical Air Hose

First notification may be noise if failure is in deck side section of umbilical. More usually Diver will report switching to emergency gas supply (bailout). A large rupture will be accompanied by the cessation in the fluctuation in demand gauge with a corresponding drop in outlet pressure. In this scenario, Diver will still have communications with the surface.

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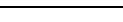
Supervisors Actions:	• IMMEDIATELY STOP all moving parts (crane / winch wires and equipment moves etc.) • Watch air pressure, be prepared to switch to secondary air (some air may be reaching Diver) • Turn on Pneumo • Direct Diver dive basket • Advise Tender to prepare recovery of Diver and umbilical • Have Standby Diver prepare to assist, if required • Be advised by Diver of the situation and react accordingly • Prepare chamber if decompression may be compromised by air shortage
Tenders Action:	• STOP all crane/winch activities • Be prepared to recover Divers umbilical • Assist Standby Diver in preparation to dive • Prepare chamber for potential omitted decompression
Divers Action:	• Switch to Bail-out Bottle and return to basket • Inform Supervisor of switch to Bail-out • Clear umbilical on return to basket • If at down line and air has not been restored advise supervisor "Diver leaving bottom" • If in basket: • Advise supervisor Diver in basket ready to leave bottom • Open basket OBG and test second stage regulator and/or hose. • Be prepared to switch to emergency air supply.

15.3.4 Contaminated Gas to Diver in Water

Under all circumstances where the purity of the Divers breathing supply is questioned, the following cause of actions must take place.

Diver reports bad tasting air supply or experiences symptoms relating to a contaminated supply.

- Dive Supervisor informed immediately by the Diver

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- Dive Supervisor switches the Diver to secondary supply
- Abort dive
- Diver taken immediately to the Dive Medic for evaluation.

15.4 Recovery of Unconscious/Injured Diver (Single Working Diver Operation)

In this situation, there is lost communication, and Tender fails to get a response from the Diver by umbilical pull signals. the following procedure shall be adopted: -

Supervisors Actions	• STOP all moving parts to seabed (crane lifts, wires, etc.) • If Diver does not respond, order the Standby Diver to assist the Diver. • Keep informed as to where the Diver is however, do not disturb the Standby Diver with too many questions. • When advised by the Standby Diver of an unconscious or injured Diver situation, prepare for a major medical emergency. • Maintain calm, even voice contact with the Standby Diver and provide appropriate advice. • Follow the Emergency Communication Procedure and request medical assistance. • Contact the onshore medical physician and have immediate transportation to the worksite provided for him if necessary. If the Diver has been injured and is losing blood, advise the medical physician of the extent of trauma and bleeding. He should bring the required plasma, Dextran, and administration sets. • Co-ordinate recovery with deck crew
Standby Divers Actions	On being directed to assist the Diver, the Standby Diver <u>MUST</u> follow the Divers umbilical from the surface On reaching the Diver the Standby Diver will: - • Check working diver condition • Open working diver free-flow valve, • If no air, close the working diver helmet free-flow valve and open working diver emergency supply;

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	- Try to get extra air to the Diver by pressing his demand valve at regular intervals; - The Diving Supervisor should have turned on the Diver's pneumo, stick it up under the neck dam of the hat (if main & standby air supplies are lost, use Standby Divers pneumo to supply stricken Diver. - Drag the Diver back dive basket checking that both umbilicals are clear. - If a dive basket is available, secure him in the basket and close the gate. Keep the Diver airway open by tilting back his head and ensure he has an air supply - Report status to surface If the Diver has returned to near normal condition (regain consciousness, talking coherently, and has good colour), take normal decompression ascent stops.
Topside Activities	- Prepare DDC to receive injured diver - Place first aid supplies, DMAC15 kit and blankets in the DDC. - Lock in DMT to attend to the Diver whilst in the DDC. - Check gas supplies and tables for Table 6 decompression profiles.
Tenders Actions	- Prepare lifting device (strop, shackles, carabiner) and crane (if available) to recover Diver from water if dive basket is unavailable. - Recover both Divers and Standby Divers umbilical as instructed by the Diving Supervisor - Recover dive basket - Clear passage between recovery site and DDC - Arrange method of transferring Diver into DDC – stretcher or blanket - If the Diver regains consciousness, beware of shock reaction and treat accordingly. - Remove the Diver's mask or helmet. DO NOT DELAY RECOMPRESSION

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Immediate First Aid in Chamber	• Check breathing and heartbeat. • If breathing, advise topside of situation. If breathing or heart has stopped, begin immediate first aid (CPR). • If the Diver is not breathing, ensure the airway is clear, tilt head back and immediately commence mouth-to-mouth resuscitation. A Plastic airway (oropharyngeal airway) in the first aid kit may be used. Continue until normal breathing has begun. While giving mouth to- mouth resuscitation, check every 30 seconds for continued heartbeat or pulse. The carotid artery (along windpipe), temporal artery (temple of head), and ulnar artery (wrist below thumb) will provide easiest pulse detection. • If the Diver's heart has stopped, it is essential that closed-chest cardiac massage be given immediately. Irreversible damage to the brain and other organs begins within four (4) minutes after the heart stops beating. If alone, give cardiac massage for 15 compressions followed by two (2) rapid mouth-to-mouth respiration's; for two (2) people the ratio is 5 and 1 instead of 15:2. Then, repeat the cycle. Cardiac activity usually begins spontaneously within 20 minutes.If the heartbeat has not begun after 20 to 30 minutes, hypoxic injury may be extensive. However, the effort to revive him should not be abandoned as he has both an elevated ppO2 and hypothermia may be in his favour. • Once the Diver has begun to breathe normally, advise the Supervisor of the situation, if this has not been done before. • If the working Diver has sustained a severe cut or amputation accompanied by extensive bleeding, it is essential that the blood loss be reduced before ascent to the surface. Apply a direct pressure bandage to the wound area or apply a light tourniquet. Do not completely stop blood flow in limb. This can lead to reduced oxygenation of the tissue and result in damage to the limb. Release tourniquet every 15 minutes for one (1) minute and then reapply. Medical supplies are in the Chamber first aid kit.
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15.5 Recovery of Unconscious/Injured Divers (Two Working Diver Operations)

In this situation, there is lost communication, and Tender fails to get a response from both divers by umbilical pull signals. the following procedure shall be adopted: -

Supervisors Actions	• STOP all moving parts to seabed (crane lifts, wires, etc.) • If Divers do not respond, order the Standby Diver to enter the water and locate the divers. • Keep informed as to where the Diver is however, do not disturb the Standby Diver with too many questions. • When advised by the Standby Diver of an unconscious or injured Diver situation, prepare for a major medical emergency. • Maintain calm, even voice contact with the Standby Diver and provide appropriate advice. • Recover the unconscious / injured diver to the basket and secure using the diver support rigging. • Ensure the Standby diver has protected the airway of the diver prior to leaving the basket in order to assist the second diver. • Keep informed as to where the Diver is however, do not disturb the Standby Diver with too many questions. • When advised by the standby diver of a second unconscious Diver inform the deck crew that this will be a two diver recovery and re compression. • Recover the unconscious / injured diver to the basket and secure using the diver support rigging. • Follow the Emergency Communication Procedure and request medical assistance. • Contact the onshore medical physician and have immediate transportation to the worksite provided for him if necessary. If the Diver has been injured and is losing blood, advise the medical physcian of the extent of trauma and bleeding. He should bring the required plasma, Dextran, and administration sets. • Co-ordinate recovery with deck crew
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Standby Divers Actions	On being directed to assist the Diver the Standby Diver <u>MUST</u> follow the Divers umbilical from the surface <u>First Diver Recovery</u> On reaching the first Diver the Standby Diver will: • Check Diver status • Open his free-flow valve, • If no air - close the free-flow valve, • Open his emergency supply. • Try to get extra air to the Diver by pressing his demand valve at regular intervals. • The Diving Supervisor should have turned on the Diver's pneumo, stick it up under the neck dam of the hat (if main & standby air supplies are lost, use Standby Divers pneumo to supply stricken Diver. • Drag the Diver back to the down line or dive basket checking that both umbilicals are clear. • If a dive basket is available secure him in the basket and close the gate. Keep the Divers airway open by tilting back his head and ensure he has an air supply. • Report status to surface • If the Diver has returned to near normal condition (regain consciousness, talking coherently, and has good color), assist him back to the basket and secure him using the diver support rigging in the basket before going to recover the second diver. <u>Second Diver Recovery</u> • Upon leaving the basket the Standby Diver must follow the diver's umbilical in order to find the second diver. • On reaching the Second Diver, the Standby Diver will: • Check Diver status. • Open his free-flow valve, • If no air - close the free-flow valve,
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	• Open his emergency supply. • Try to get extra air to the Diver by pressing his demand valve at regular intervals. • The Diving Supervisor should have turned on the Diver's Pneumo, stick it up under the neck dam of the hat (if main & standby air supplies are lost, use Standby Divers Pneumo to supply stricken Diver. • Drag the Diver back to the down line or dive basket checking that both umbilicals are clear. • Secure him in the basket and close the gate. Keep the Divers airway open by tilting back his head and ensure he has an air supply. • Report status to surface Ensure all three umbilicals are clear to the surface and report status to the Supervisor
Topside Activities	• Prepare DDC to receive injured diver. • Place first aid supplies, DMAC15 kit and blankets in the DDC. • Lock in DMT to attend to the Diver whilst in the DDC. • Check gas supplies and tables for Table 6 decompression profiles.
Tenders Actions	• Prepare lifting device (strop, shackles, carabiner) and crane (if available) to recover Diver from water if dive basket is unavailable. • Recover both Divers and Standby Divers umbilical as instructed by the Diving Supervisor • Recover dive basket (if available) • Clear passage between recovery site and DDC • Arrange method of transferring Diver into DDC – stretcher, one will have to be provided by vessel. • If the Diver regains consciousness, beware of shock and treat accordingly. • Remove the Diver's mask or helmet. DO NOT DELAY RECOMPRESSION

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Immediate First Aid in Chamber	• Check breathing and heartbeat. • If breathing, advise topside of situation. If breathing or heart has stopped, begin immediate first aid (CPR). • If the Diver is not breathing, ensure the airway is clear, tilt head back and immediately commence mouth-to-mouth resuscitation. A Plastic airway (oropharyngeal airway) in the DMAC 15 may be used. Continue until normal breathing has begun. While giving mouth to-mouth resuscitation, check every 30 seconds for continued heartbeat or pulse. The carotid artery (along windpipe), temporal artery (temple of head), and ulnar artery (wrist below thumb) will provide easiest pulse detection. • If the Diver's heart has stopped, it is essential that closed-chest cardiac massage be given immediately. Irreversible damage to the brain and other organs begins within four (4) minutes after the heart stops beating. If alone, give cardiac massage for 30 compressions followed by two (2) rapid mouth-to-mouth respirations. Then, repeat the cycle. If an automatic or semi- automatic defibrillator is available place this on him after the first cardiac massage cycle, then continue as instructed by the Defibrillator. Cardiac activity usually begins spontaneously within 20 minutes. If the heartbeat has not begun after 20 to 30 minutes, hypoxic injury may be extensive. However, the effort to revive him should not be abandoned as he has both an elevated ppO2 and hypothermia may be in his favor. • Once the Diver has begun to breathe normally, advise the Supervisor of the situation, if this has not been done before. • If the working Diver has sustained a severe cut or amputation accompanied by extensive bleeding, it is essential that the blood loss be reduced before ascent to the surface. Apply a direct pressure bandage to the wound area or apply a light tourniquet. Release tourniquet every 15 minutes for one (1) minute and then reapply. Medical supplies are in the chamber first aid Kit and the DMAC 15 Kit
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15.6 Fouled/Trapped Diver

Surface Indications	- Diver informs Supervisor. - Surface Tender informs supervisor
Subsea Indications	- Diver fouled – Can't move. - Possible loss of umbilical service - Diver being pulled off job

Surface Actions	- All stop vessel and equipment moves. - Supervisor informs bridge and deck crew. - The supervisor instructs ROV to monitor. - Supervisor instructs tender to recover slack umbilical. - Recover diver if decompression schedule has been compromised. - Carry out decompression if required on recovery of diver
Subsea actions	- Diver attempts to clear umbilical. - Diver requests assistance from another diver or surface diver - Diver keeps supervisor informed. - Check umbilical for damage

15.7 Severed Diver Umbilical - Loss of Communication and Air Supply

This is a serious and potentially life-threatening occurrence. It is likely that there will be an immediate loss of communications, air to the Diver, and power to hat lights and video. At night this can be extremely disorientating to the Diver in addition to his loss of air. Response must be both calm and swift.

The complete severance of the umbilical may result in the Diver being unable to negotiate back to the down line or basket, especially if visibility is impaired. Or if a significant section of umbilical remains attached to the Diver, it may impede recovery of the Diver.

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Supervisors Actions:	• Remain calm. • IMMEDIATELY STOP all moving parts • Watch air pressure, be prepared to switch to secondary air (some air may be reaching Diver) • Turn on pneumo • Notify deck crew of the situation. • Advise Tender to give one (1) pull on the umbilical to ascertain Diver status. • Give full brief on known status Standby Diver. • Jump Standby Diver • Prepare chamber for decompression for potential Therapeutic Table
Tenders Action:	• STOP all crane/winch activities. • Be prepared to recover Diver's umbilical. • Assist Standby Diver in preparation to dive. • Prepare chamber for appropriate treatment table. • At night send cyalume sticks down the down line or dive basket wire(provides a reference as well as illumination for Diver).
Divers Actions:	• Remain Calm (particularly at night) • Switch to bail out. • Assess location in relationship to down line or dive basket. • At night, if disorientated and unable to locate down line / basket, WAIT for Standby Diver • Assess whether umbilical will impede ascent to surface. • If possible, go to the diving basket and use emergency air supply.WAIT for Standby Diver • If no basket is available go to down line and WAIT for Standby Diver • REMEMBER Standby Diver will be following Diver's umbilical
Standby Divers Actions:	• Carry sharp knife and adjustable spanner (and torch if at night) • Follow Divers Umbilical from Surface

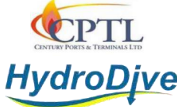
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	• On reaching Diver; Calm Diver • Check Divers Bail Out Bottle gauge for contents. • Report immediate status to Supervisor. • Assess and clear any umbilical fouling that will prevent Diver recovery. • Advise Supervisor of impediment to Diver recovery • Be prepared to use pneumo hose to supplement Diver's air. • Assist recovery of Diver up down line (or suitable means of controlled ascent) or direct the recovery of Diver basket. • Be prepared to enter chamber for Omitted Decompression
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15.8 DP Degraded Status – Amber

Surface Indications	• DP Light/Alarm activated. • Informed by DP Operator • Thrusters ramp up. • Vessel may lean over
Subsea Indications	• Informed by supervisor. • Diver pulled from worksite. • Noise of thrusters

Surface actions	• Supervisors inform diver to move to safe location usually return to LARS. • Assess reason for the degraded DP status. • If safe to continue, continue with dive. • If not safe to continue dive, recover Diver and umbilical to surface. • Investigate reason for reason for the degraded DP status
Subsea actions	• Divers move to safe location or return to LARS. • Diver informs supervisor when at safe location and all slack umbilical recovered • Diver await further instructions from supervisor to abort dive or continue dive

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15.9 DP Loss of Position – RED

Surface Indications	• DP Light/Alarm activated • Informed by DP Operator • Thrusters ramp up • Vessel may lean over
Subsea Indications	• Informed by supervisor • Diver pulled from worksite • Noise of thrusters

Surface actions	• Supervisor abort dive and Instruct divers to return to the LARS immediately • DP operator to use all available means to limit vessel loss of position • Recover Diver and umbilical to surface • Investigate reason for loss of vessel position
Subsea actions	• Divers return to LARS immediately for recovery • Diver coil umbilical over arm • Diver informs supervisor as soon as umbilical is clear of seabed obstructions and is inside LARS

15.10 Evacuation of Divers Undergoing Decompression Treatment in Chamber

In the event that the chamber and/or chamber occupants are endangered, the following abort procedures should be followed to evacuate the divers according to the following levels of emergency.

- Level 1 Abort – Where time is available to complete the decompression procedure.
- Level 2 Abort – Where time is not available due to impending natural disasters or mechanical failures

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The medical physician should be consulted at the earliest opportunity. All personnel should be made aware that the possibility of decompression illness complication is high, in addition consideration should be given that the attendant (inside the chamber) might be suffering of decompression illness due to the procedure.

15.10.1 Level 1 Abort Procedure

The air decompression schedule used in recompression treatment aborts is modified by having all chamber occupants begin breathing oxygen as soon as a depth of 30ft or shallower is reached.

Oxygen breathing periods of 25 minutes on oxygen, followed by 5 minutes on air are continued until the total time on oxygen is one half or more of the total decompression time required by the selected table. Upon surfacing, chamber occupants are treated as if they had surfaced from a normal dive.

15.10.2 Level 2 Abort Procedure

Where there is an imminent danger to the chamber occupants such as fire, explosion or possible sinking of the installation that completion of the abort procedure is impossible, the following may be carried out:

- If deeper than 60ft, decompress the chamber to 60ft in one minute
- Put all the divers on pure oxygen on BIBS. Oxygen breathing is to be continued until the surface is reached without interruptions
- The chamber rate of ascent is 10ft per minute. This corresponds to 6 minutes to decompress the chamber from 60ft to surface

Upon reaching surface:

- Evacuate all chamber occupants to the nearest recompression facility. If air transport is necessary to transport divers, the cabin "altitude" should not exceed 1000' (300m)
- The divers must breathe oxygen during the surface transfer: Two oxygen resuscitators complete with cylinder fully charged with medical quality oxygen should be available for use on site

Once arrived at the recompression chamber:

- Treat symptoms of decompression illness in accordance with decompression treatment procedures
- If no symptoms occurred after the treatment was aborted, apply treatment table USN 7 to all participants as a preventive measure

During transit to the hyperbaric treatment facility the use of pain-relieving gas must not be administered.

During the planning stage of any operation which may require the implementation of these procedures, a risk assessment must be performed. The results of this must clearly define, and show to be available, the means to evacuate divers to the nearest recompression facility (s).

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15.11 Diver Adrift on the Surface

A diver adrift on the surface will have cut his umbilical in an emergency, or had it cut accidentally. He may have had to jettison his helmet and bail-out bottle and may be injured.

This situation shall be treated like any man overboard situation, with all applicable precautions. In addition to man overboard there is an element of urgency as the diver may be suffering from hyperbaric trauma. A man shall keep the diver visible and throw a life ring if possible. Fast Rescue Craft shall be launched and attempt to recover the stricken diver is safe and possible.

If he cannot be recovered immediately, the Diving Supervisor must alert the installation's safety boat and standby vessel. In a large field, there may be in-field helicopters which could also assist.

Members of the diving team should keep the drifting diver in sight for as long as possible and note his direction of drift. It can be extremely difficult to locate a person in the water in even a moderate swell.

15.12 Launch and Recovery System (LARS) Failure

If the primary LARS fails, assess the situation, and advise the diver that the planned alternate method of recovery shall be implemented;

- If the problem can be rectified without placing the diver in danger, rectify the problem and continue diving.
- If not, implement the alternate means of diver recovery (clump winch) and retrieve the diver to the surface.

In the event of complete LARS failure, where both forms of redundancy have failed (main winch & clump winch);

- the secondary LARS and standby Diver is to be launched to recover working diver(s).
- The Tender will be required to disconnect Diver 1 umbilical if 'clipped' as a means of excursion restriction.
- Both Divers will then be recovered by means of the secondary LARS. Any decompression that is required will be conducted in- water from the LARS.

16 DIVING EMERGENCY REPORTING

All accidents, incidents and near miss occurrences will be reported using the Accident & Incident Procedures (HDG-HSE-PR-001)